

BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



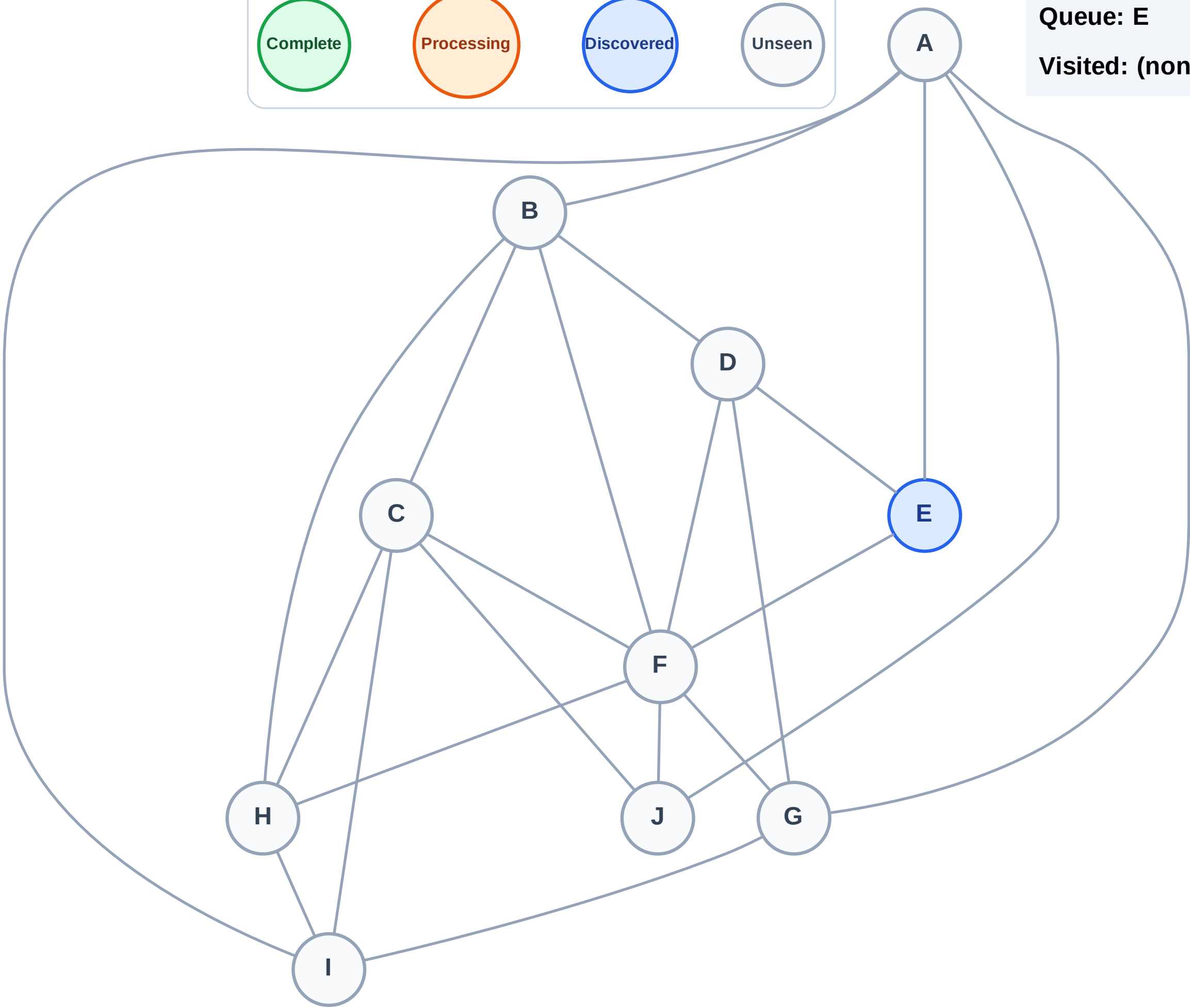
Discovered



Unseen

Queue: E

Visited: (none)



BFS Traversal

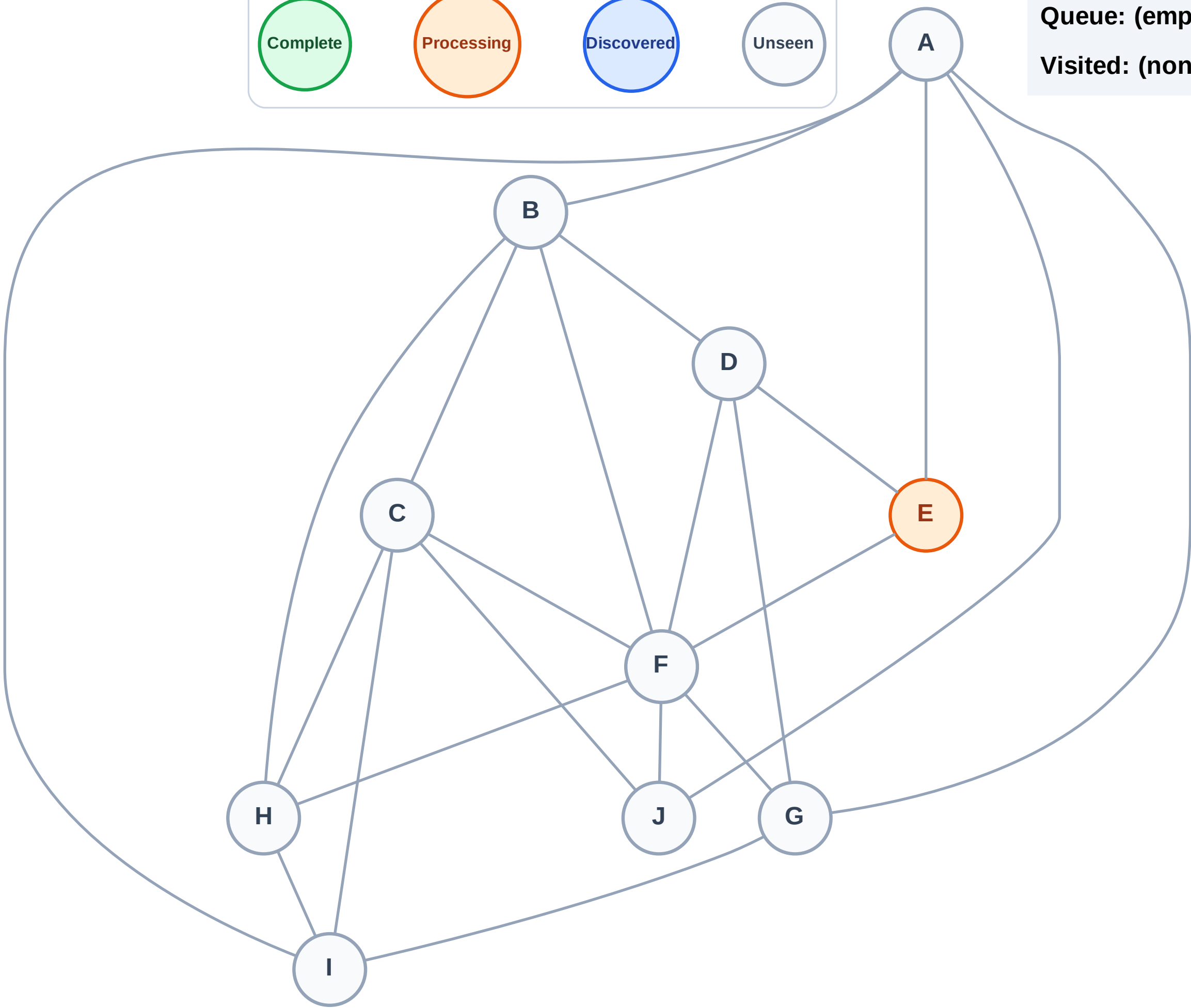
Step 2: Process node E

Legend



Queue: (empty)

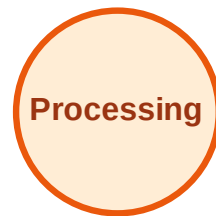
Visited: (none)



BFS Traversal

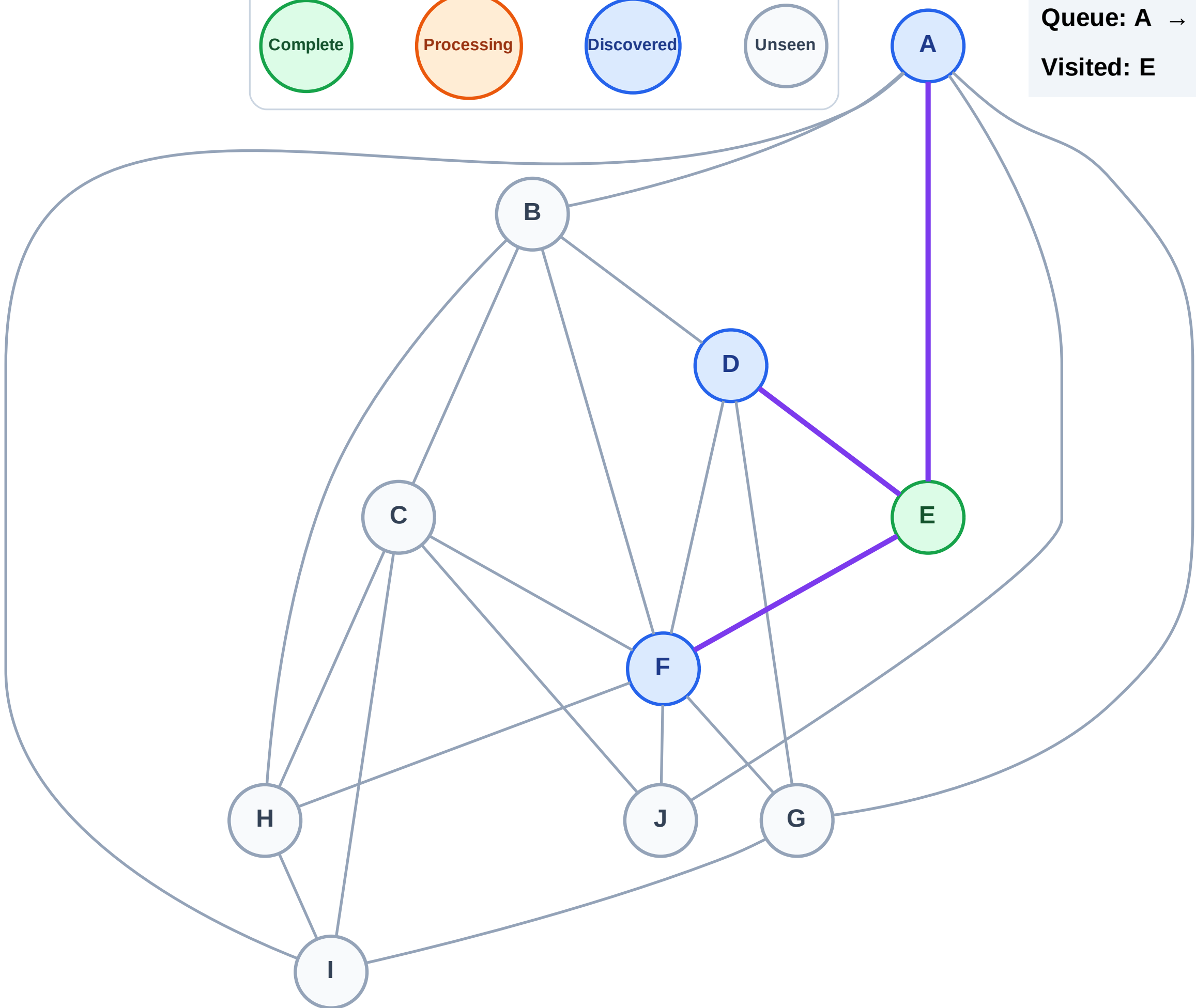
Step 3: Discover A, D, F from E

Legend



Queue: A → D → F

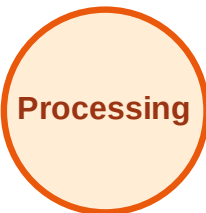
Visited: E



BFS Traversal

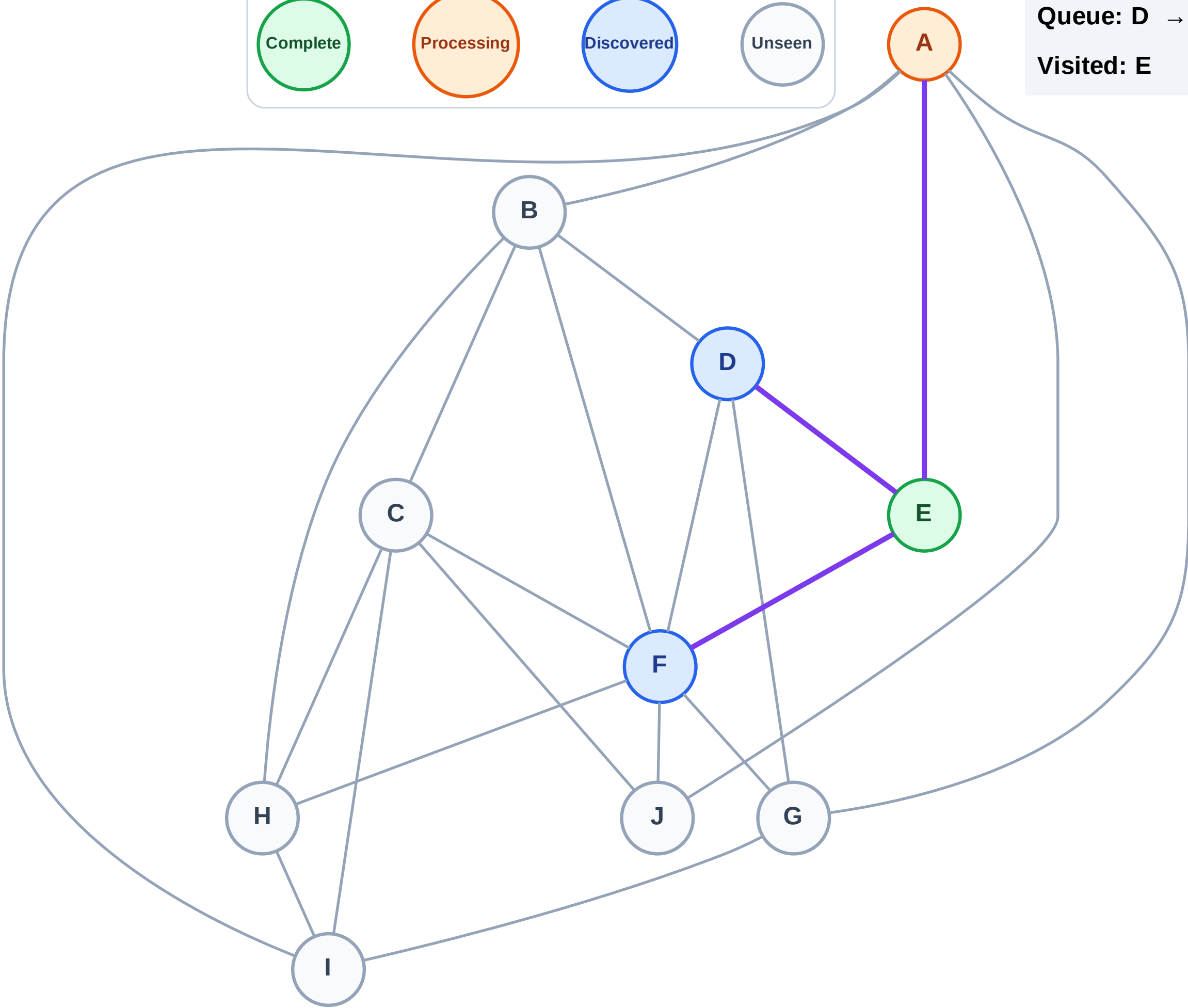
Step 4: Process node A

Legend



Queue: D → F

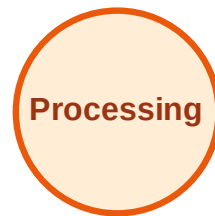
Visited: E



BFS Traversal

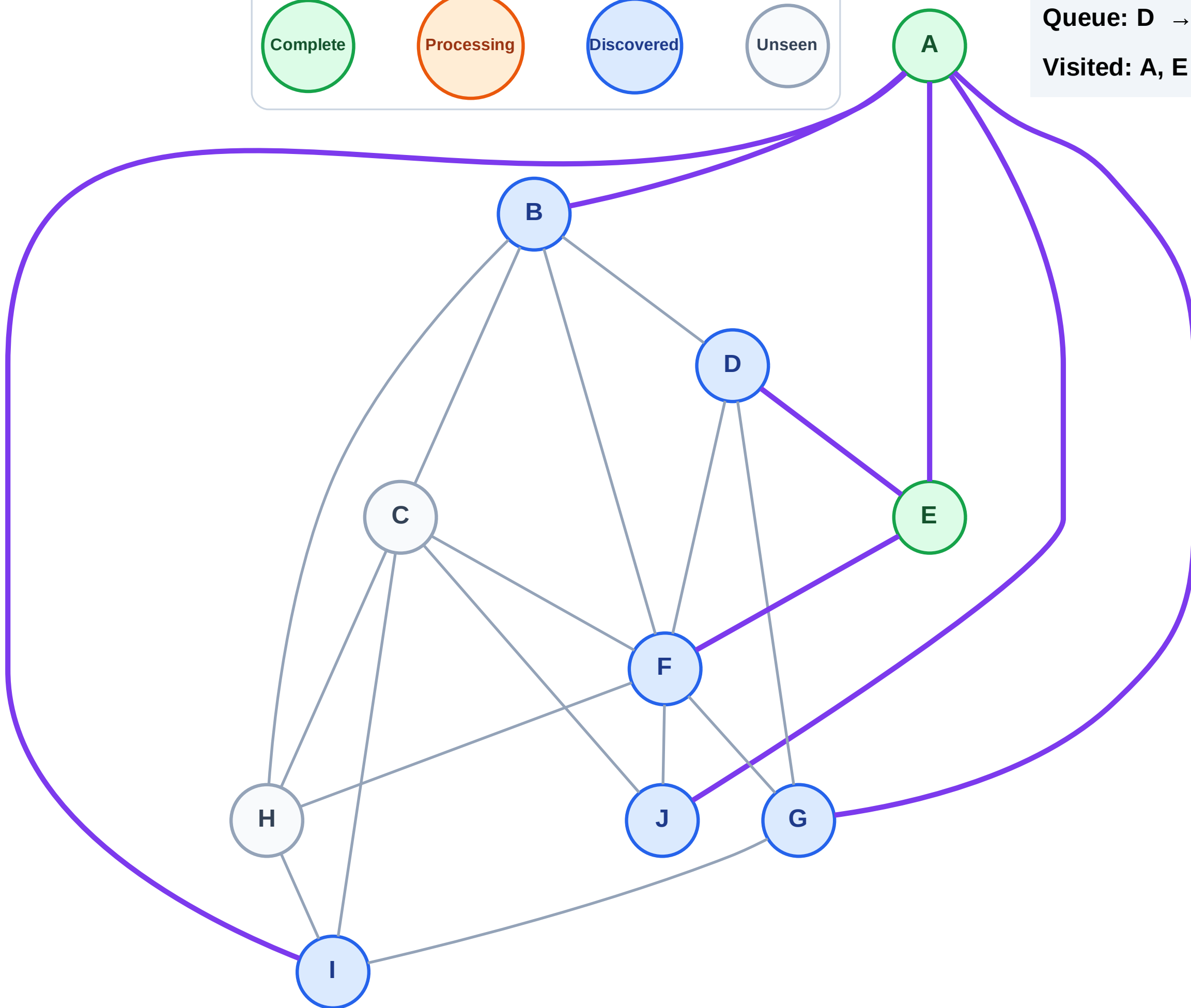
Step 5: Discover B, G, I, J from A

Legend



Queue: D → F → B → G → I → J

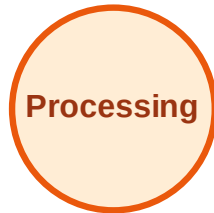
Visited: A, E



BFS Traversal

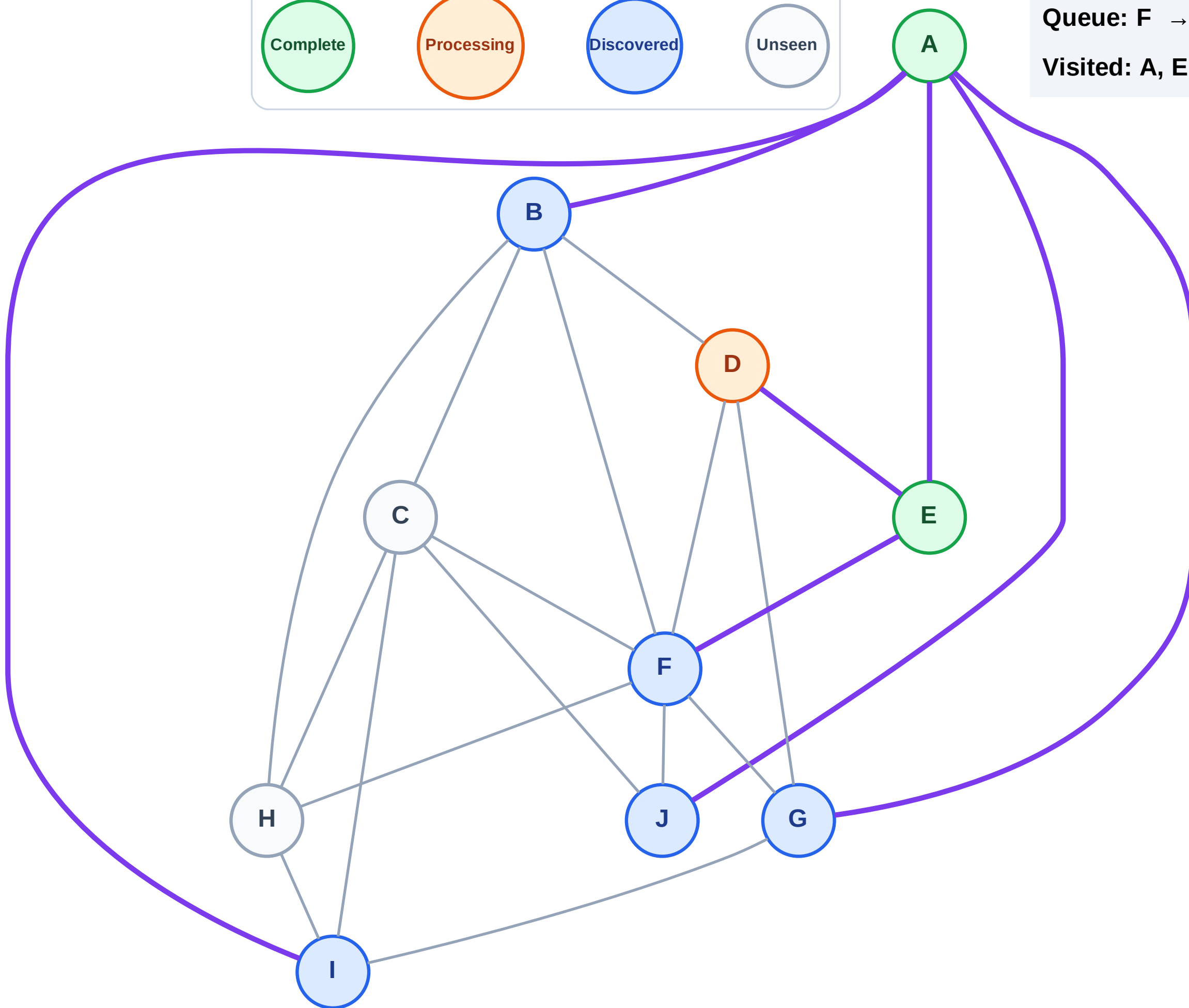
Step 6: Process node D

Legend



Queue: F → B → G → I → J

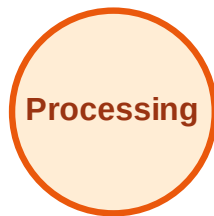
Visited: A, E



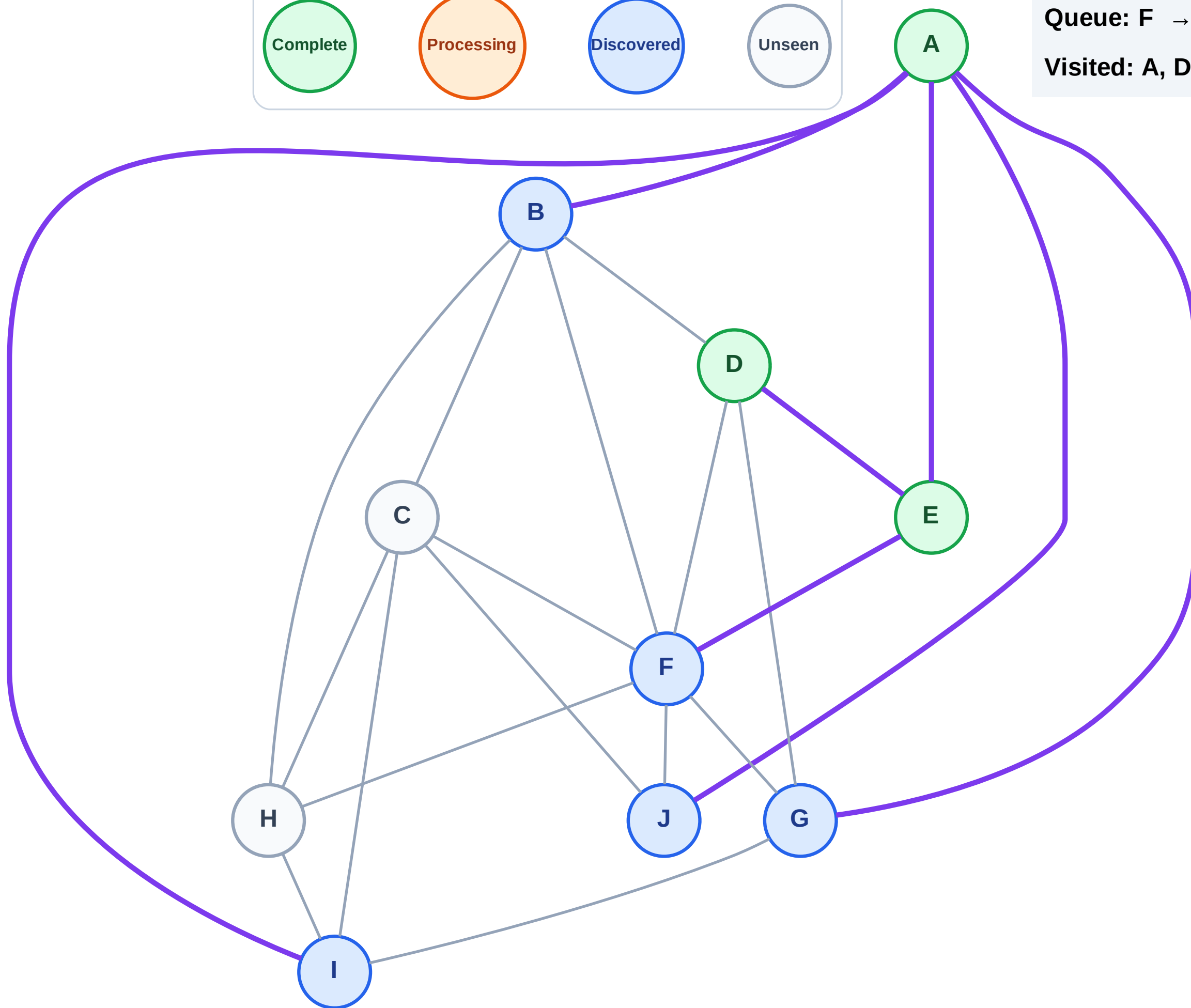
BFS Traversal

Step 7: No new neighbors from D

Legend



Queue: F → B → G → I → J
Visited: A, D, E



BFS Traversal

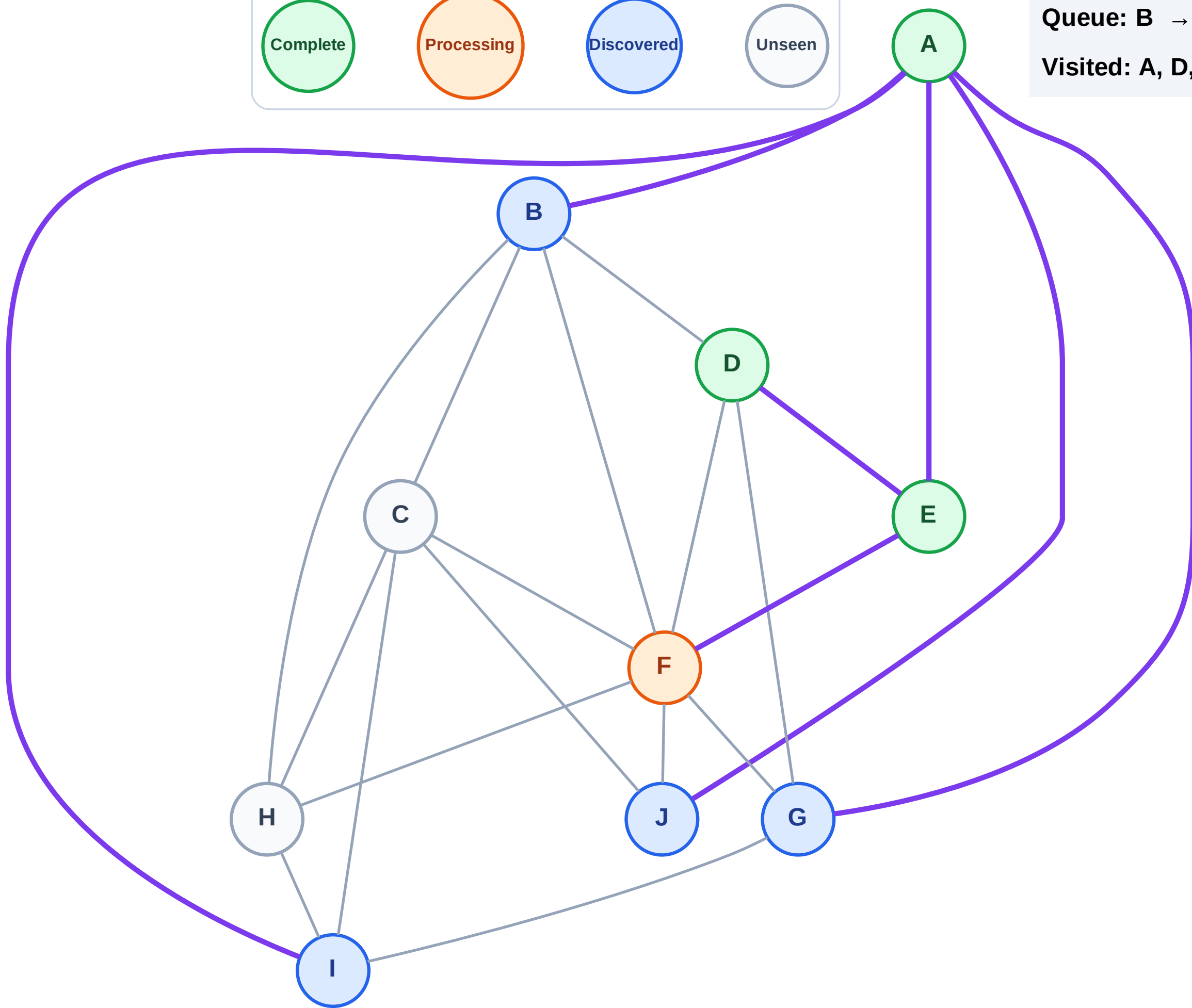
Step 8: Process node F

Legend



Queue: B → G → I → J

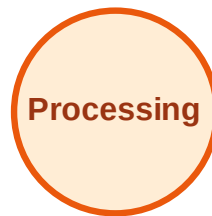
Visited: A, D, E



BFS Traversal

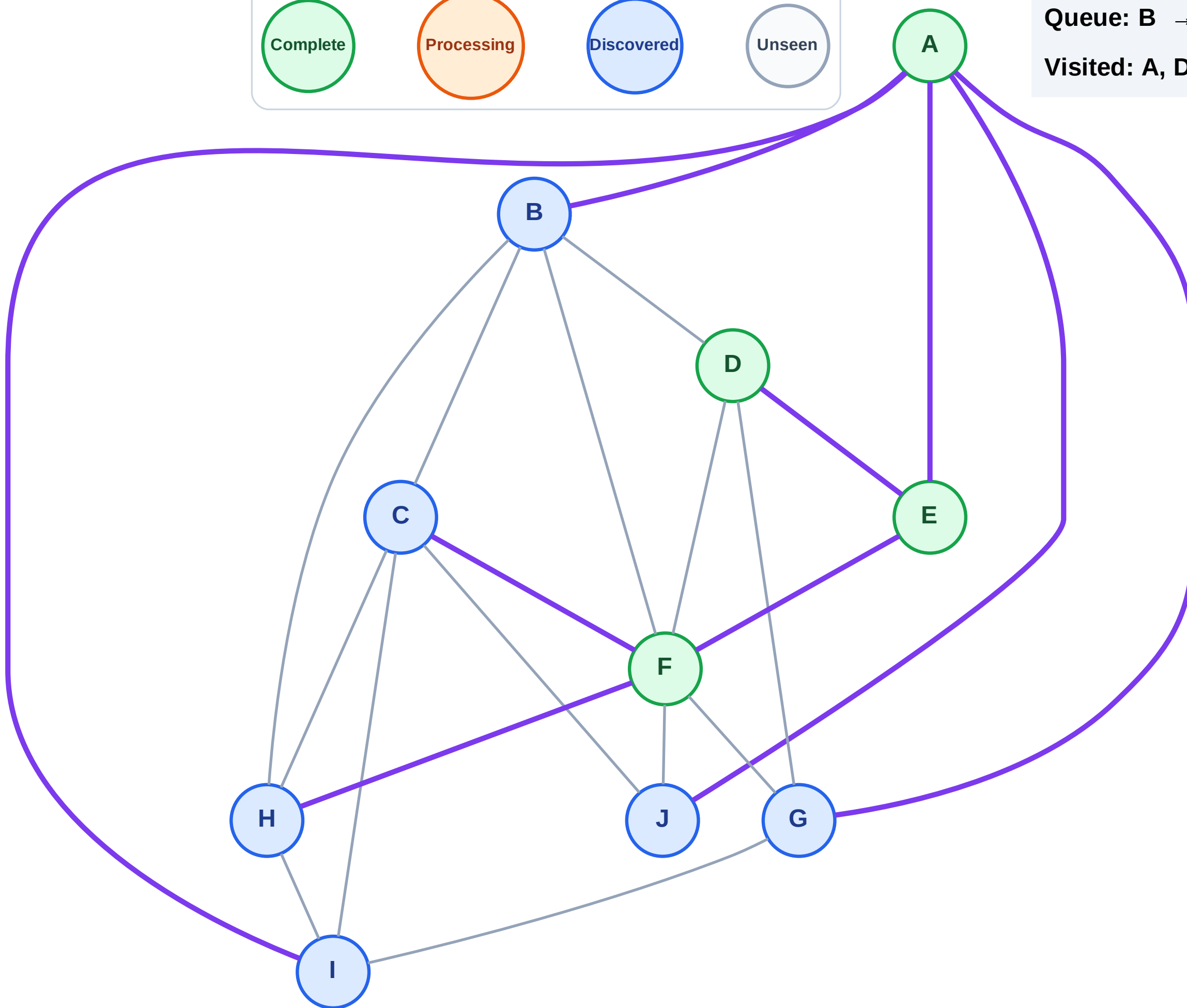
Step 9: Discover C, H from F

Legend



Queue: B → G → I → J → C → H

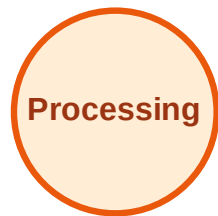
Visited: A, D, E, F



BFS Traversal

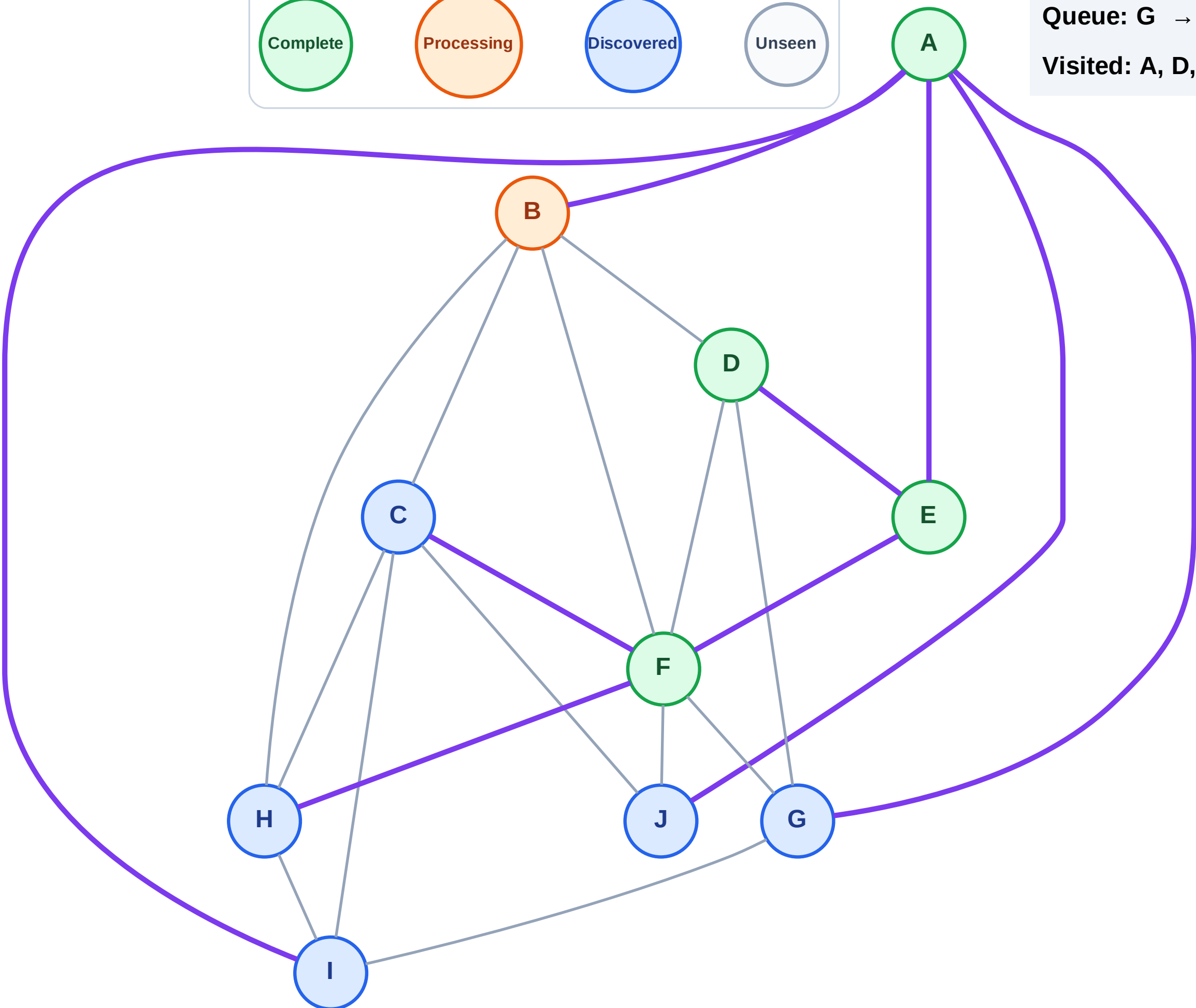
Step 10: Process node B

Legend



Queue: G → I → J → C → H

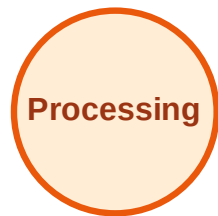
Visited: A, D, E, F



BFS Traversal

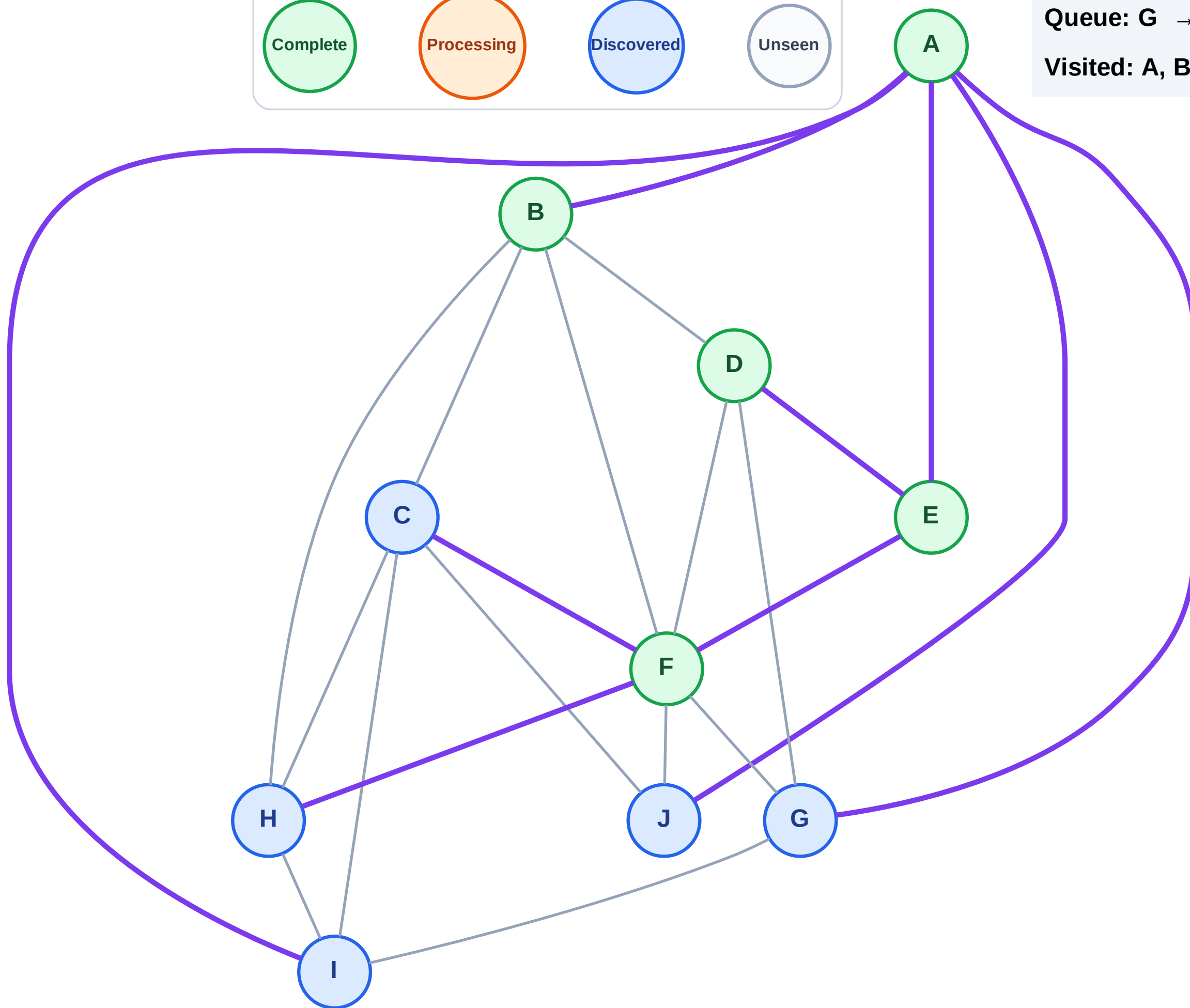
Step 11: No new neighbors from B

Legend



Queue: G → I → J → C → H

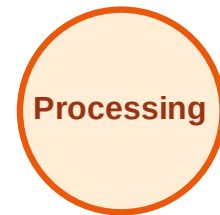
Visited: A, B, D, E, F



BFS Traversal

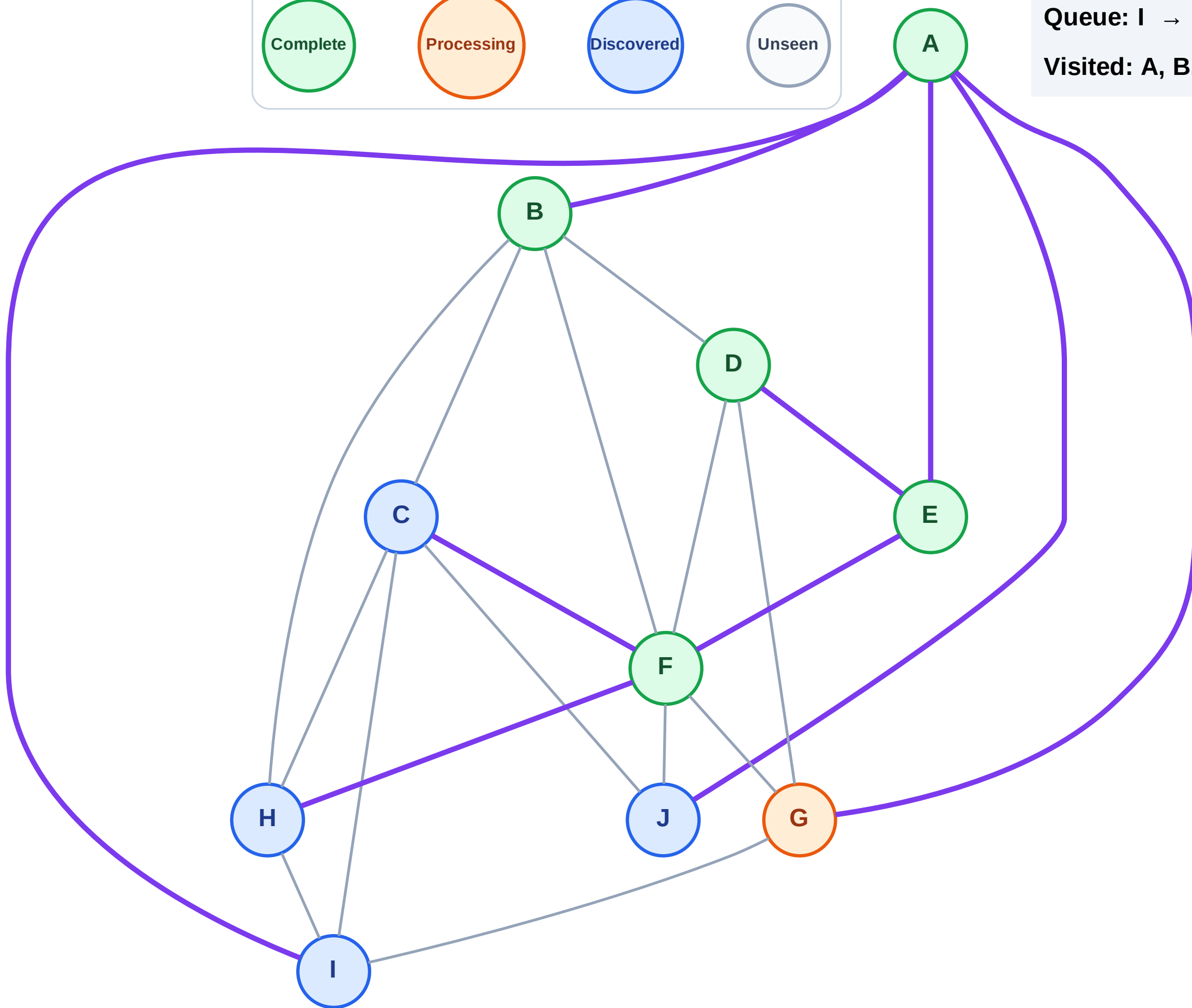
Step 12: Process node G

Legend



Queue: I → J → C → H

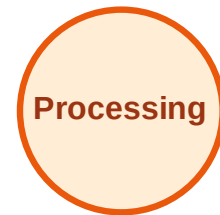
Visited: A, B, D, E, F



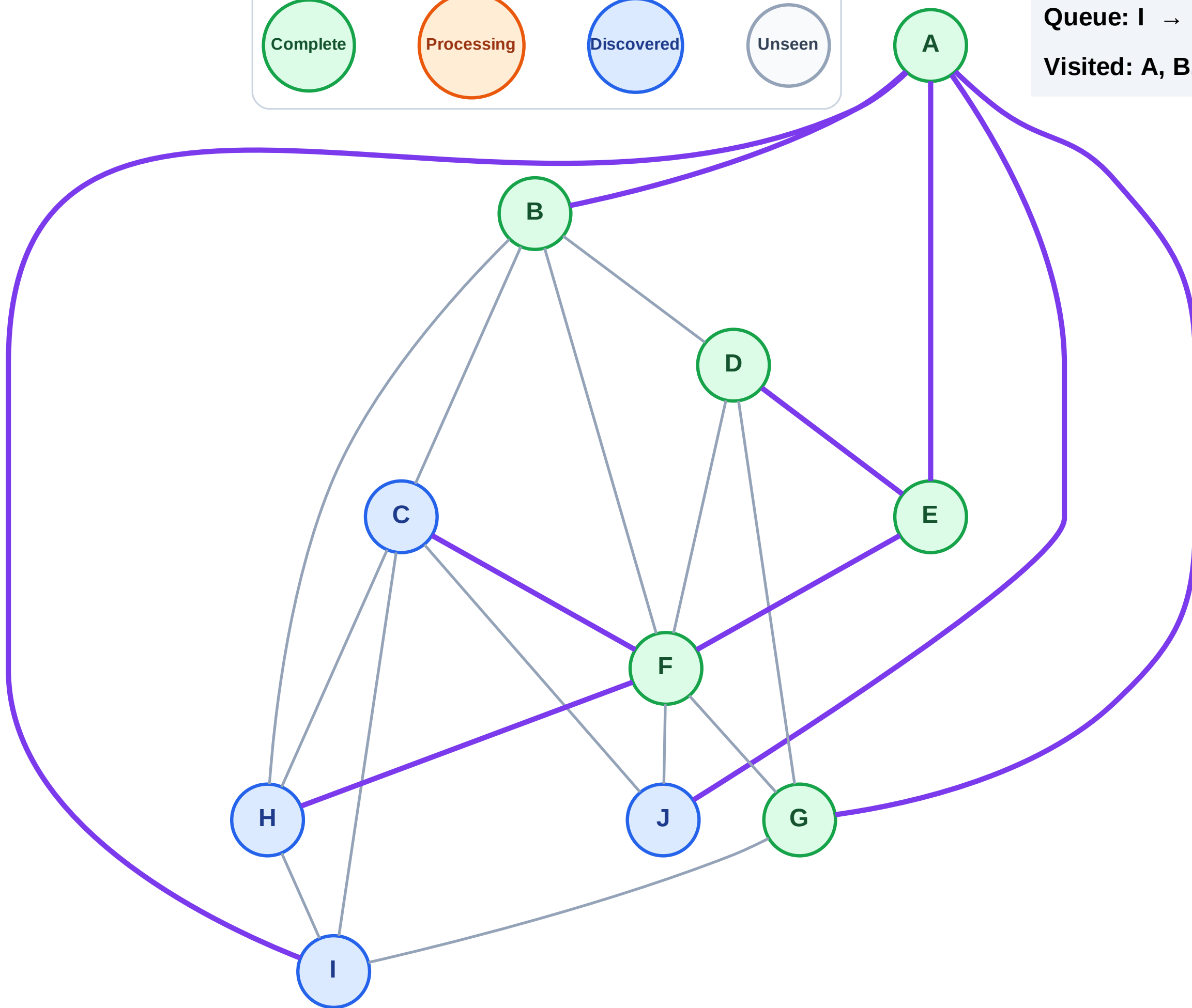
BFS Traversal

Step 13: No new neighbors from G

Legend



Queue: I → J → C → H
Visited: A, B, D, E, F, G



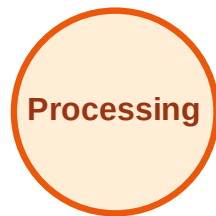
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



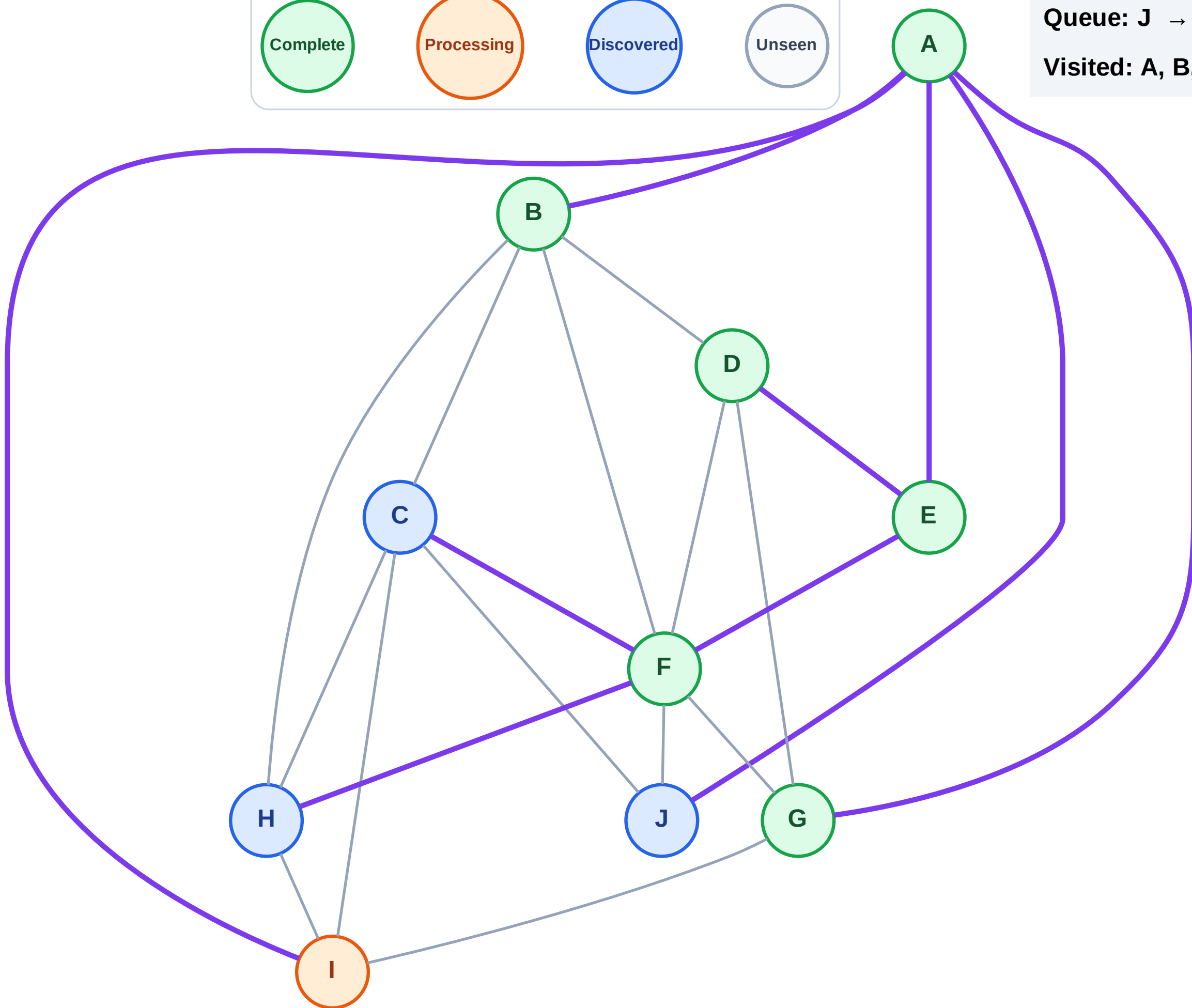
Discovered



Unseen

Queue: J → C → H

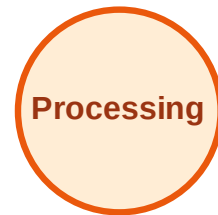
Visited: A, B, D, E, F, G



BFS Traversal

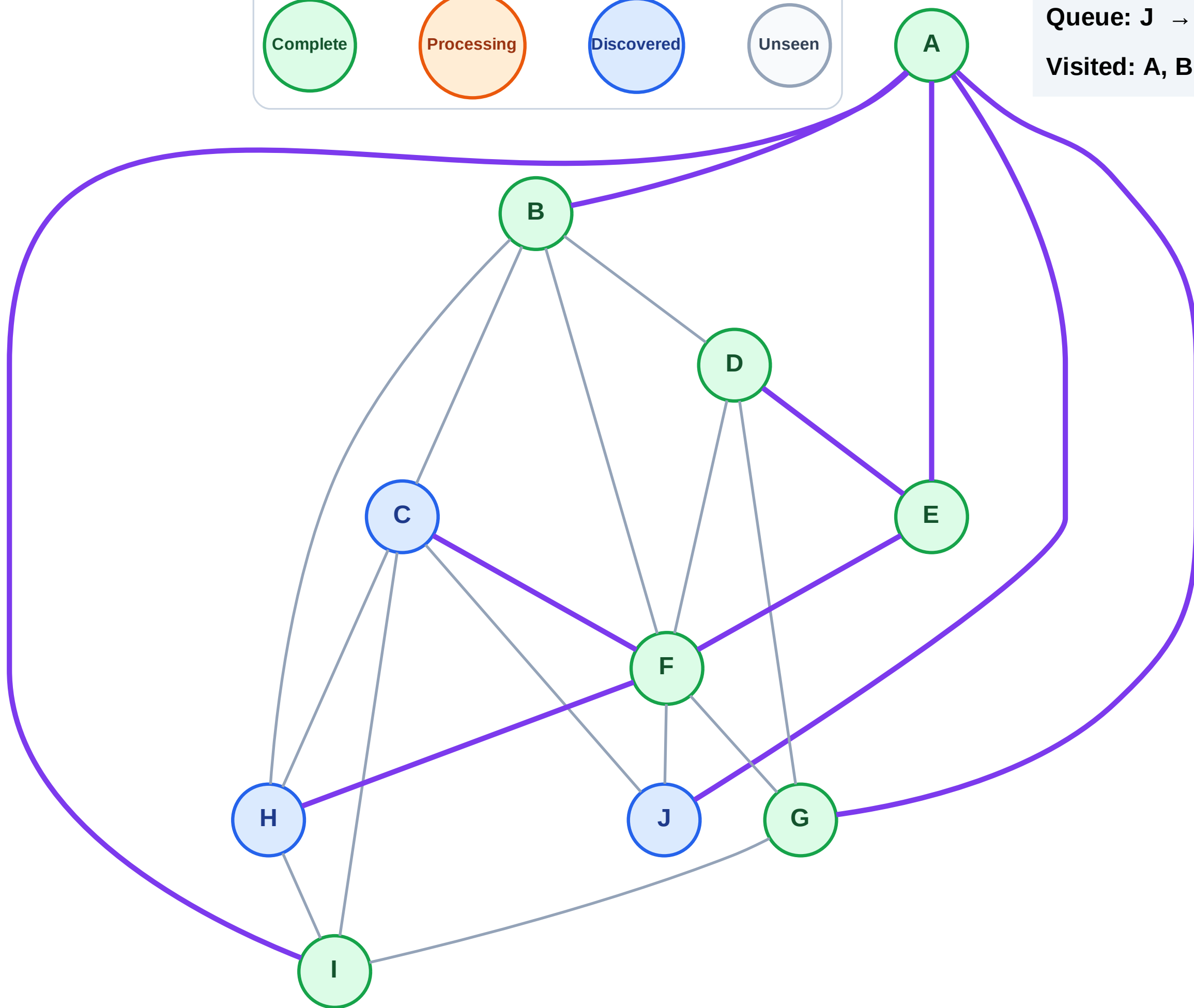
Step 15: No new neighbors from I

Legend



Queue: J → C → H

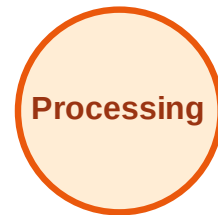
Visited: A, B, D, E, F, G, I



BFS Traversal

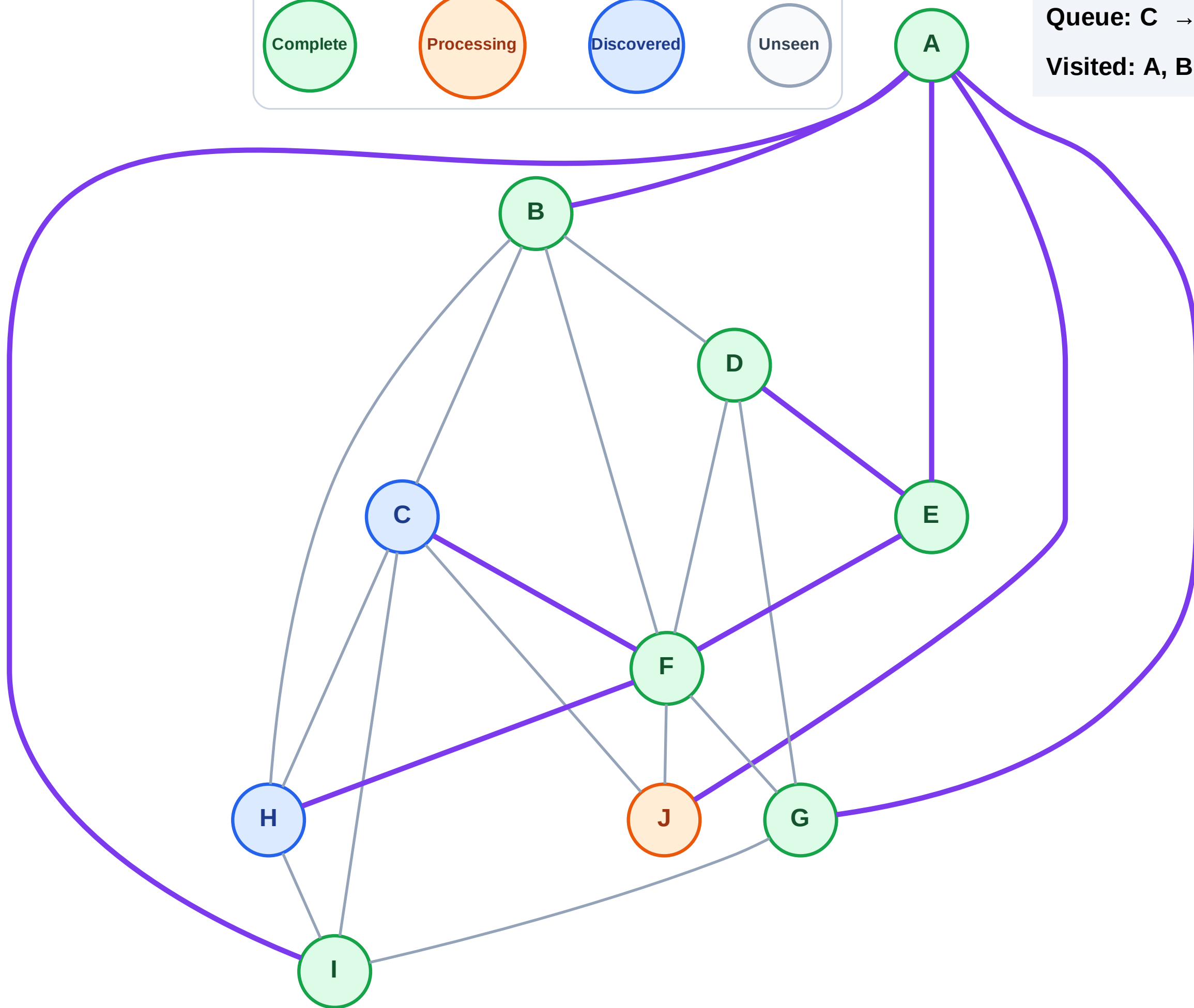
Step 16: Process node J

Legend



Queue: C → H

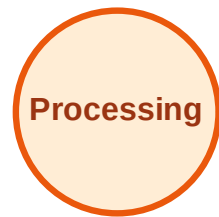
Visited: A, B, D, E, F, G, I



BFS Traversal

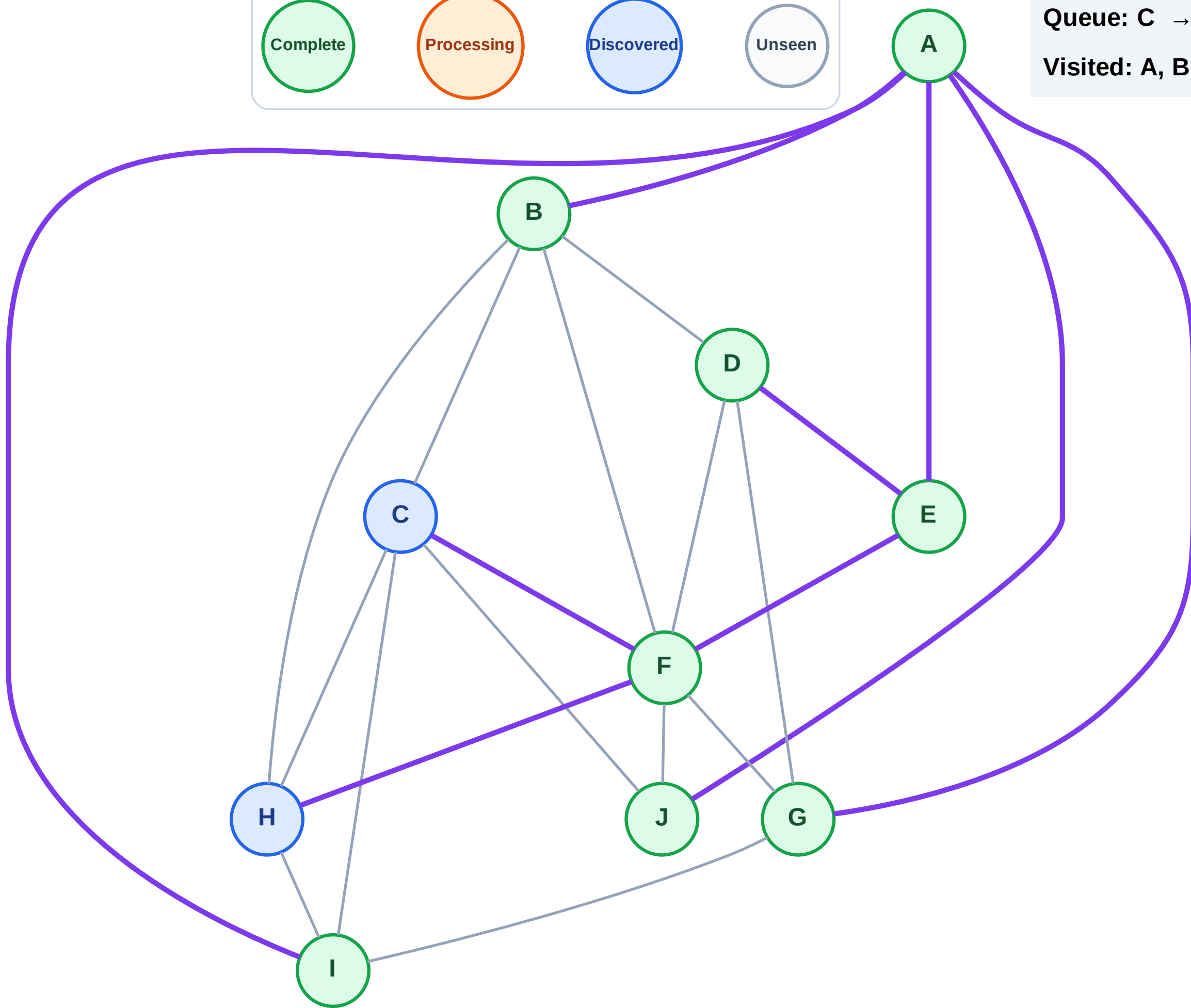
Step 17: No new neighbors from J

Legend



Queue: C → H

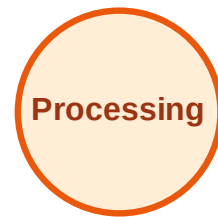
Visited: A, B, D, E, F, G, I, J



BFS Traversal

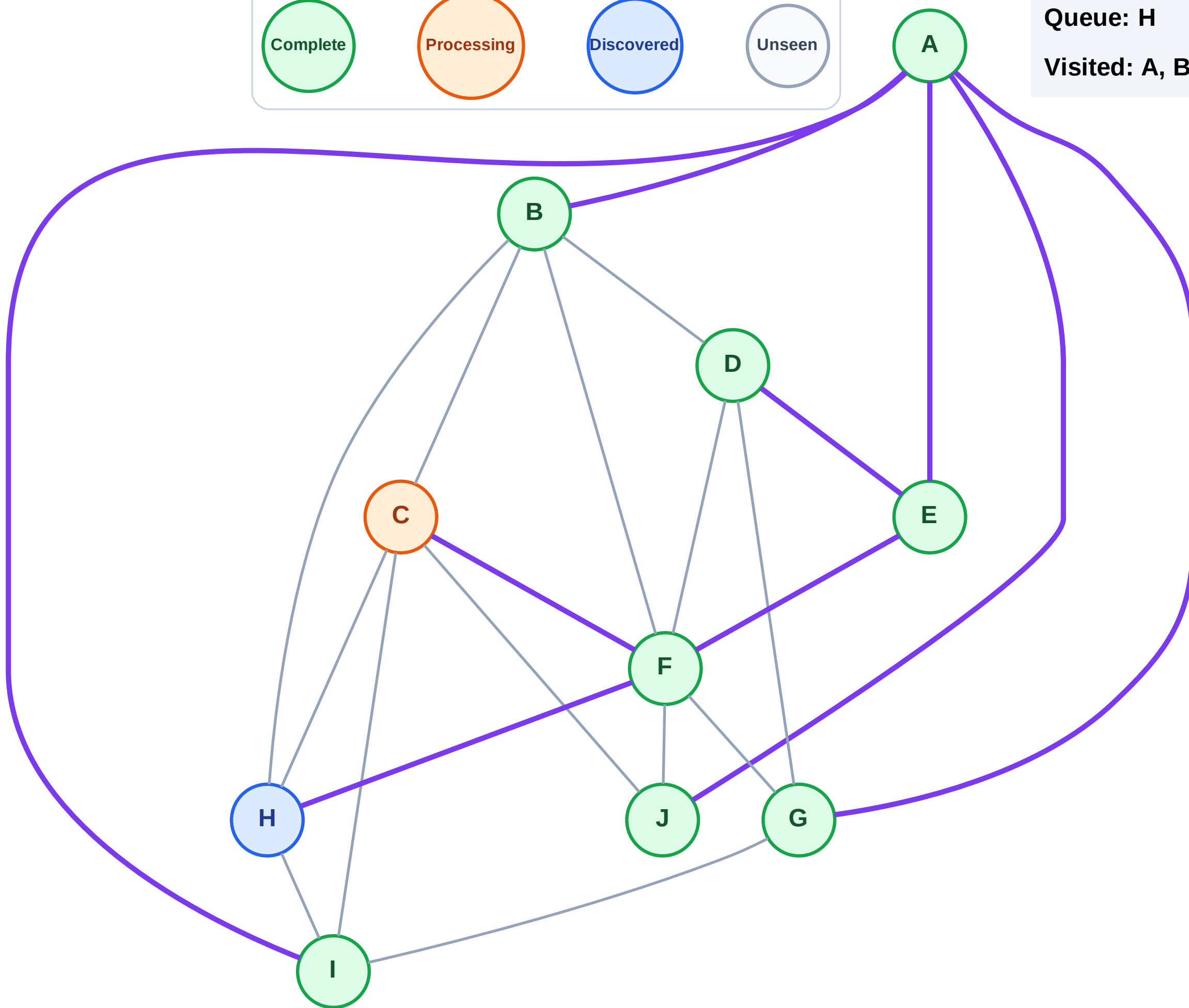
Step 18: Process node C

Legend



Queue: H

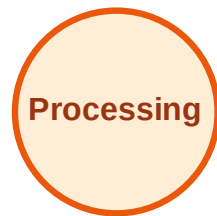
Visited: A, B, D, E, F, G, I, J



BFS Traversal

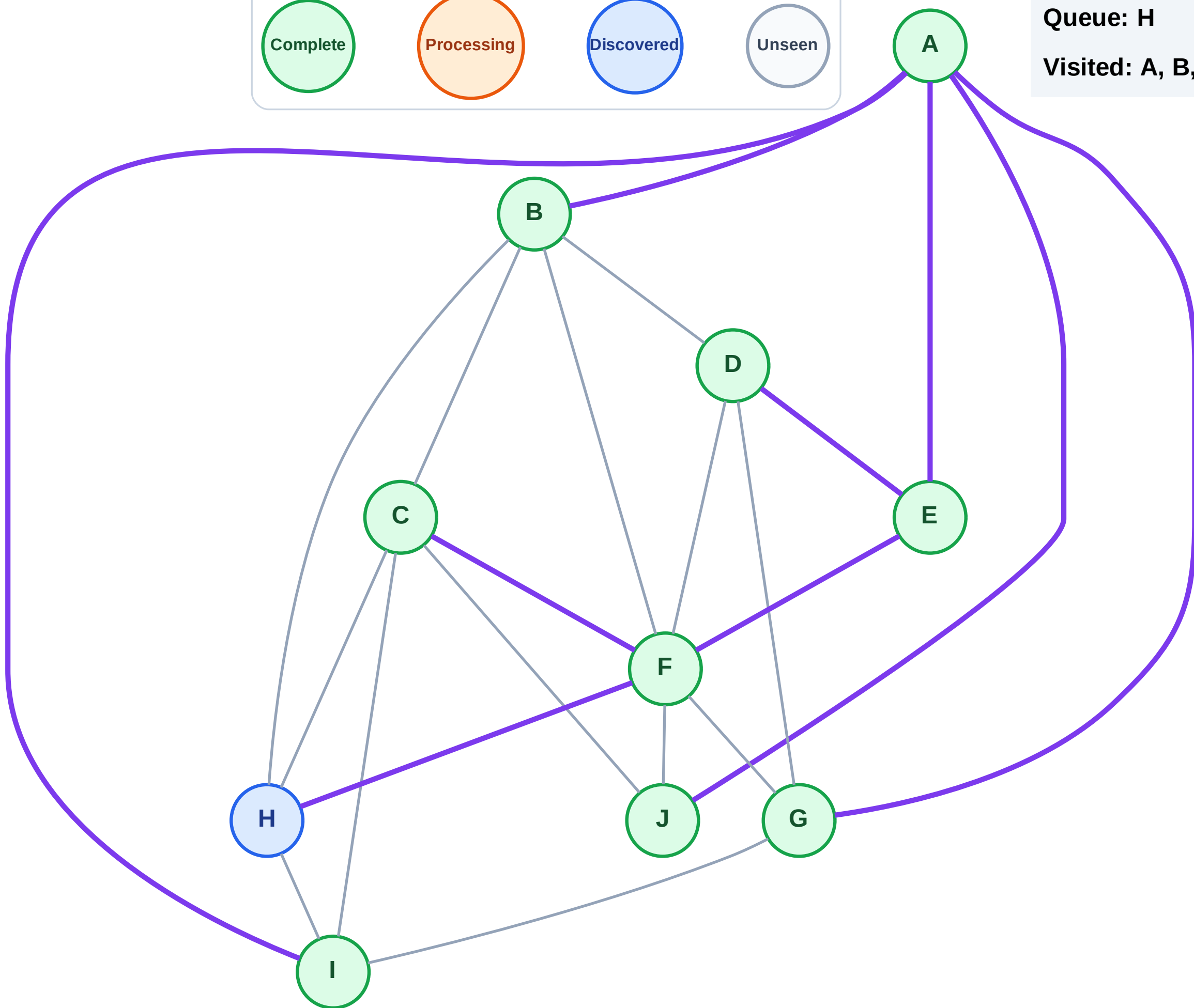
Step 19: No new neighbors from C

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



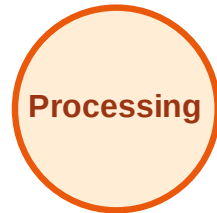
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



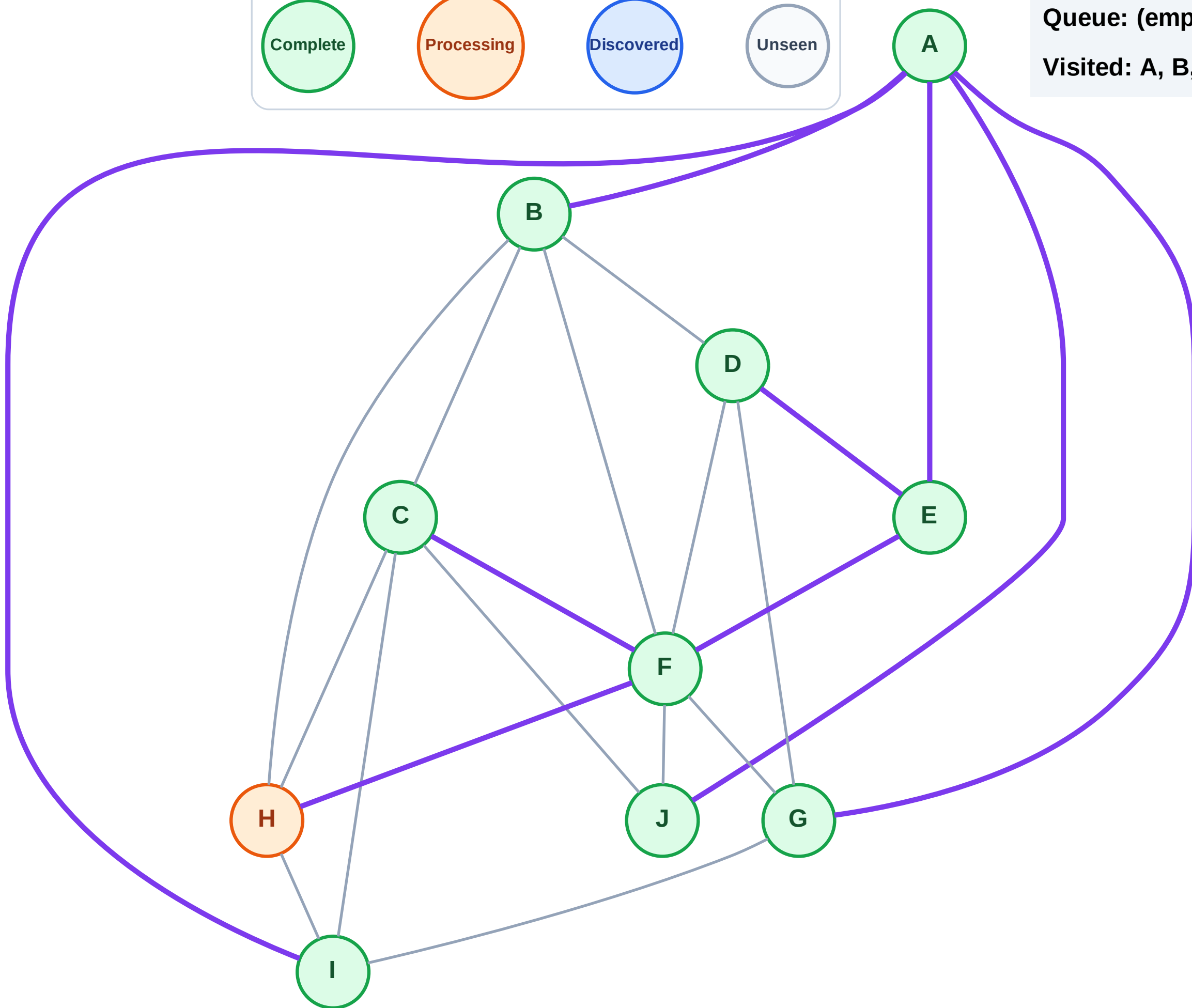
Discovered



Unseen

Queue: (empty)

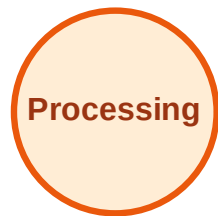
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

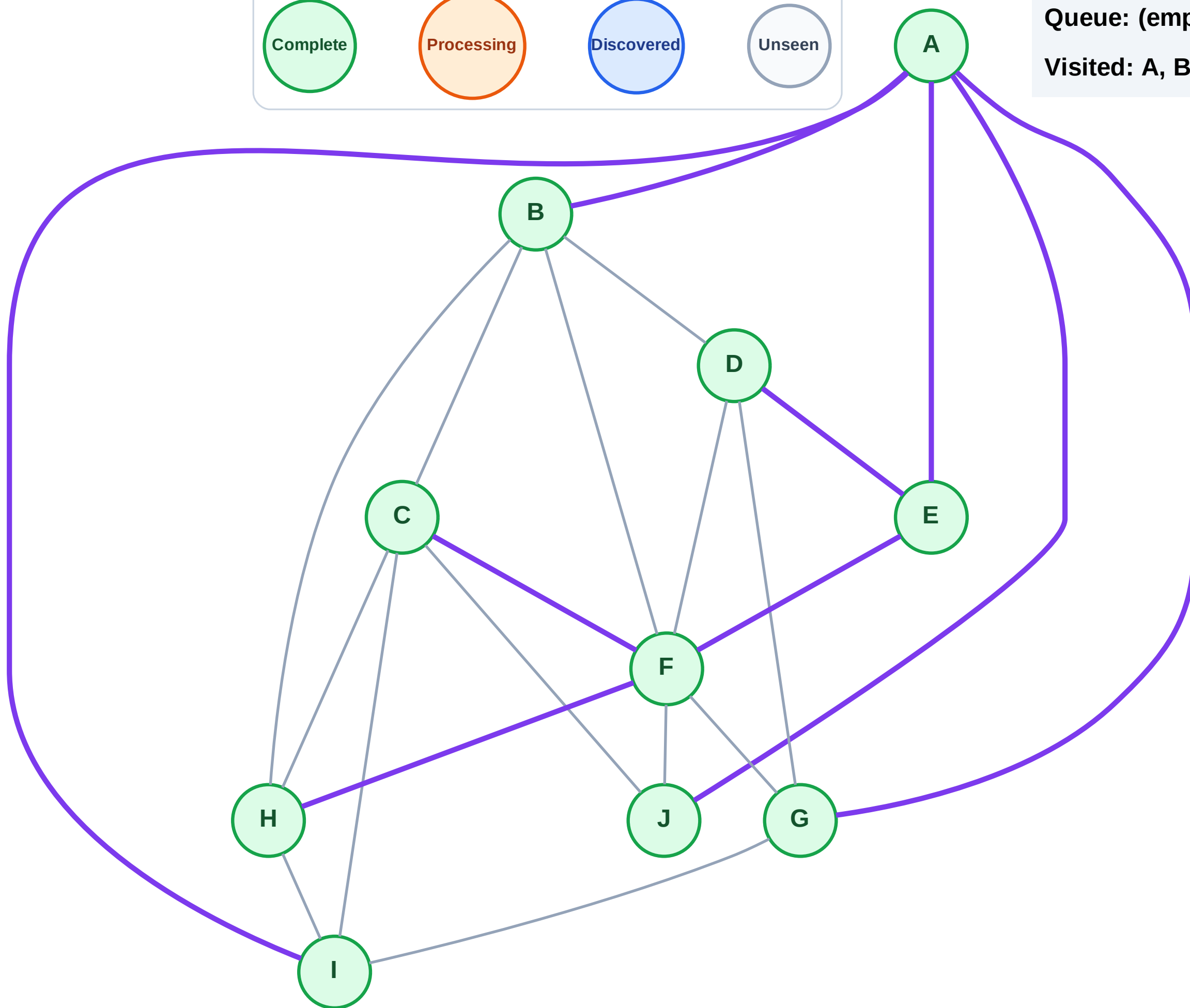
Step 21: No new neighbors from H

Legend



Queue: (empty)

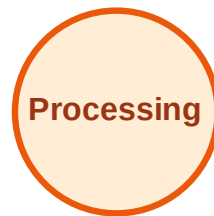
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

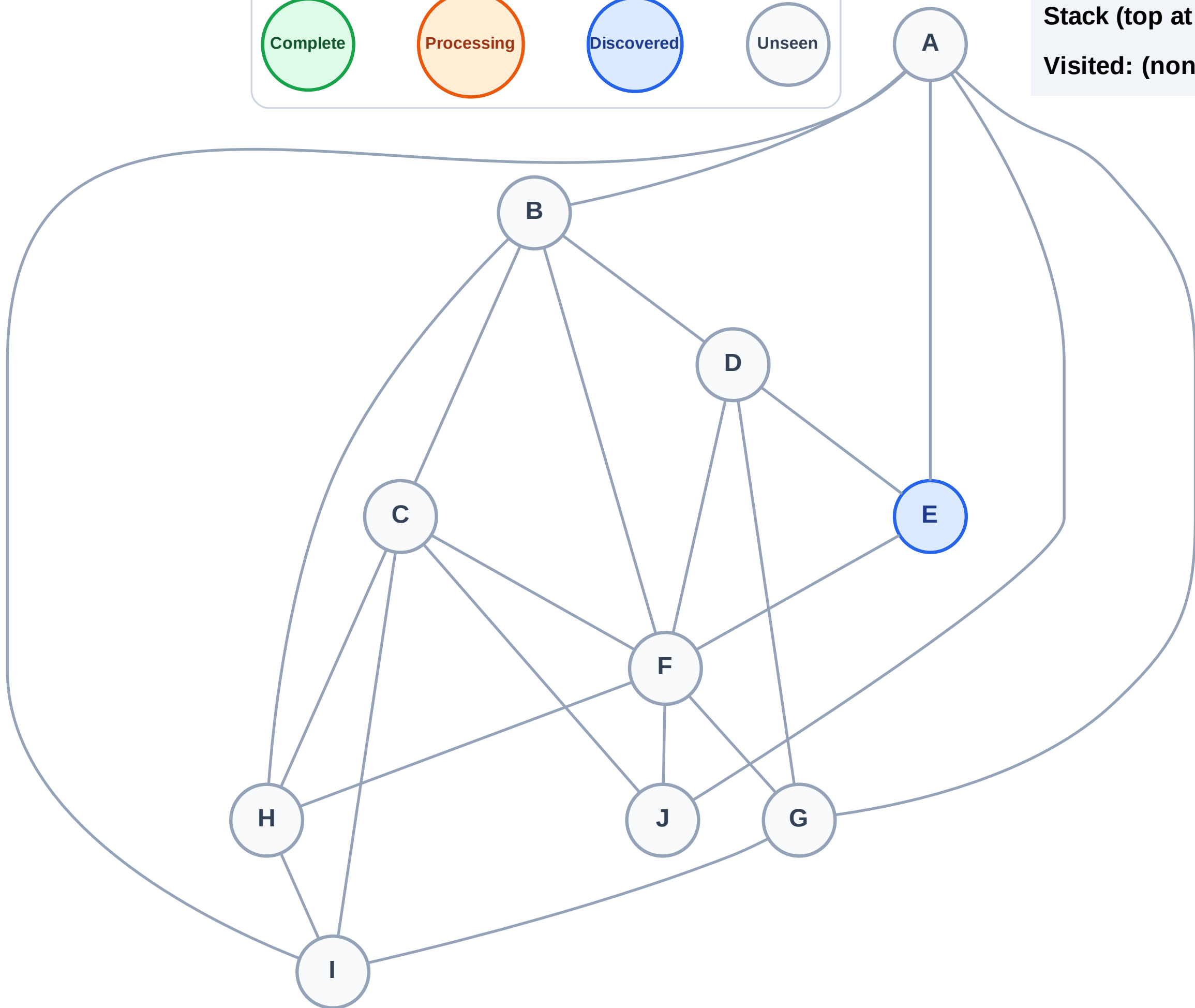
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

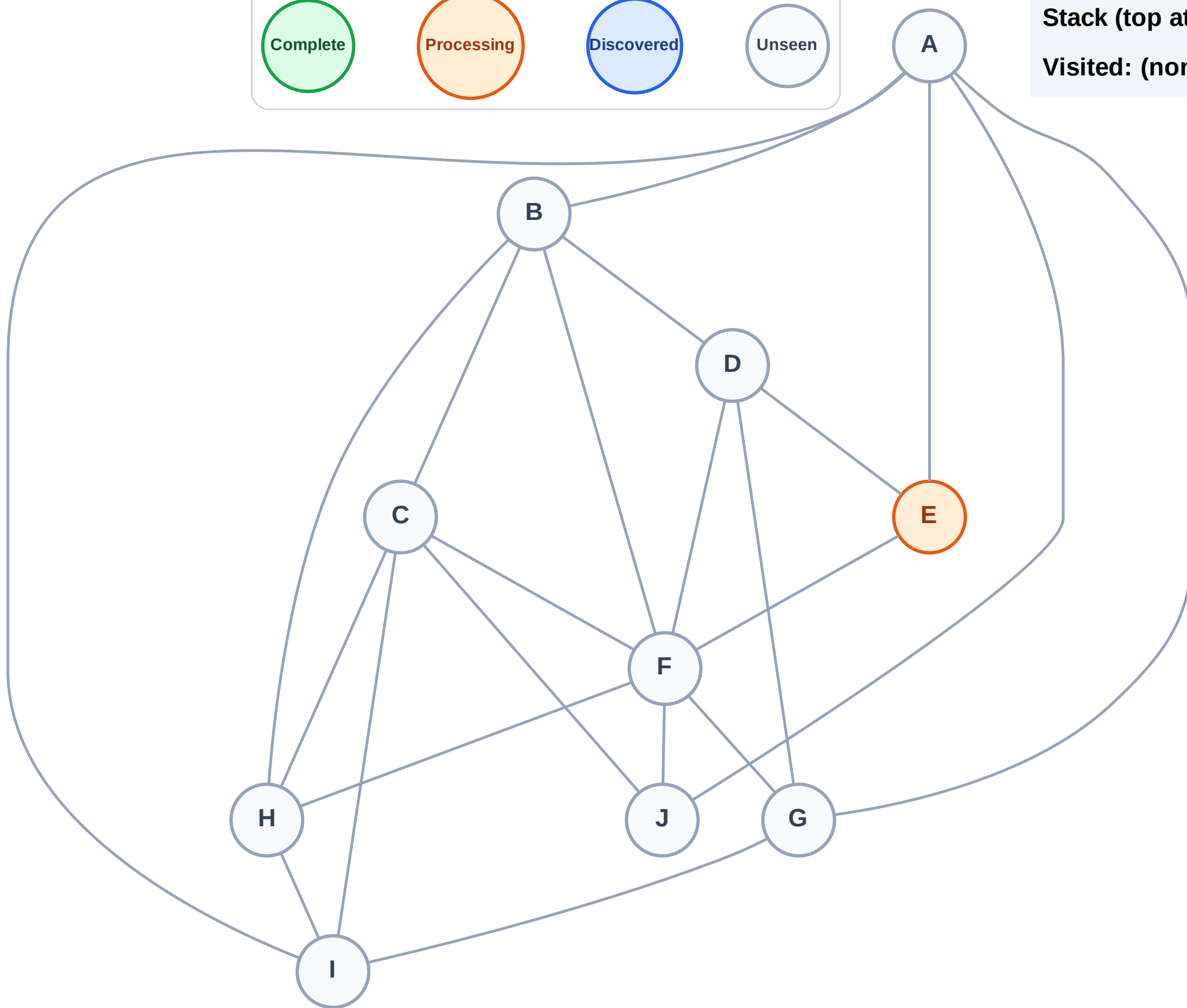
Step 2: Process node E

Legend



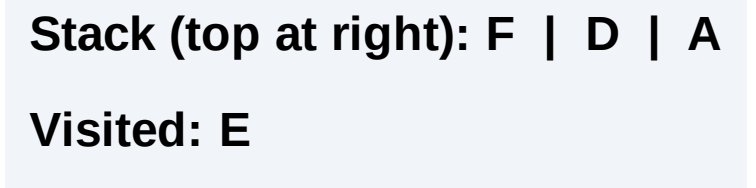
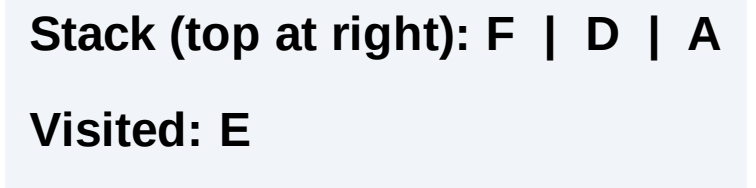
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover F, D, A from E

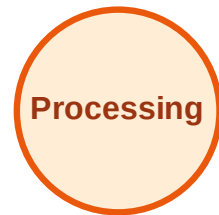
Legend



DFS Traversal

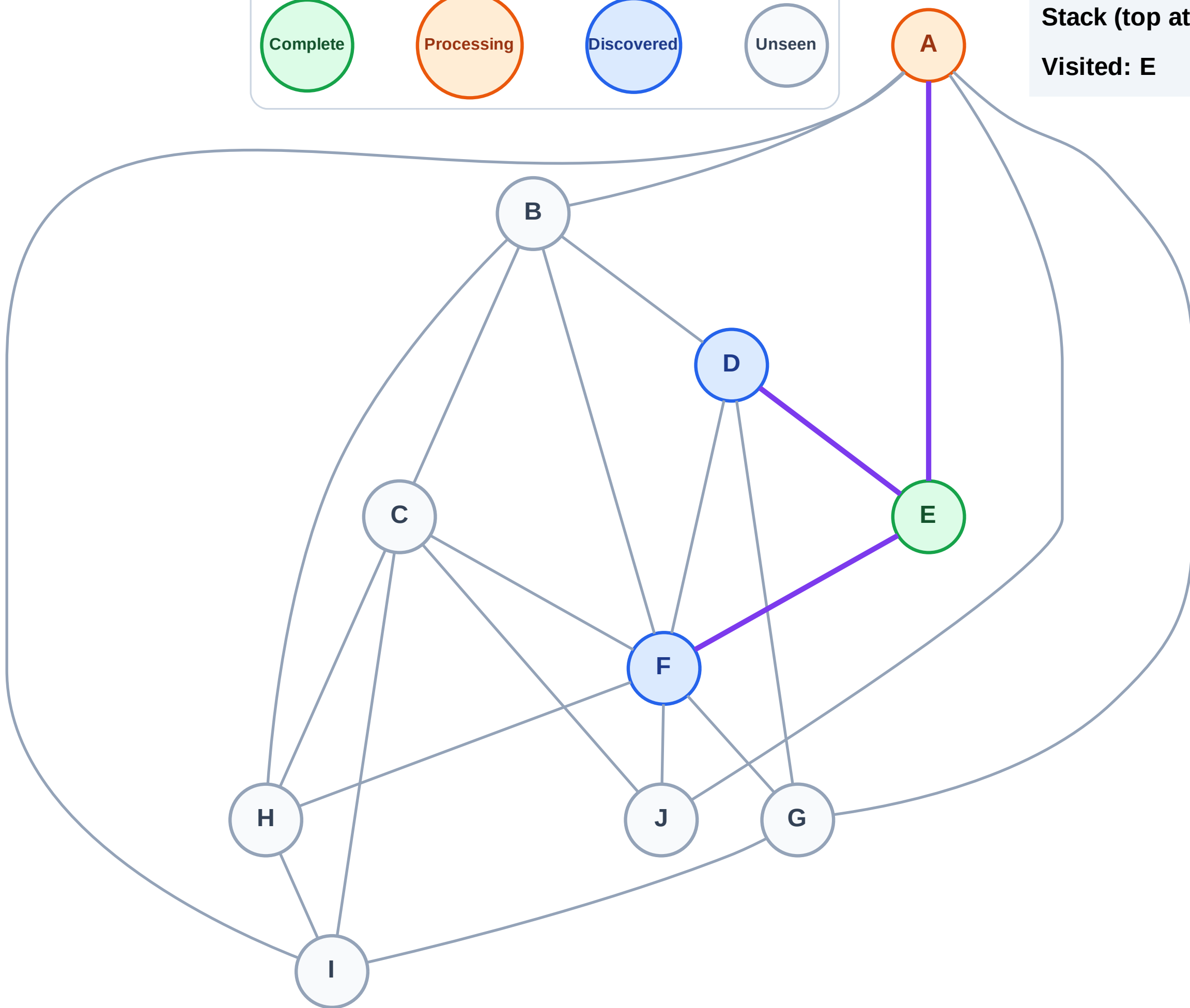
Step 4: Process node A

Legend



Stack (top at right): F | D

Visited: E



DFS Traversal

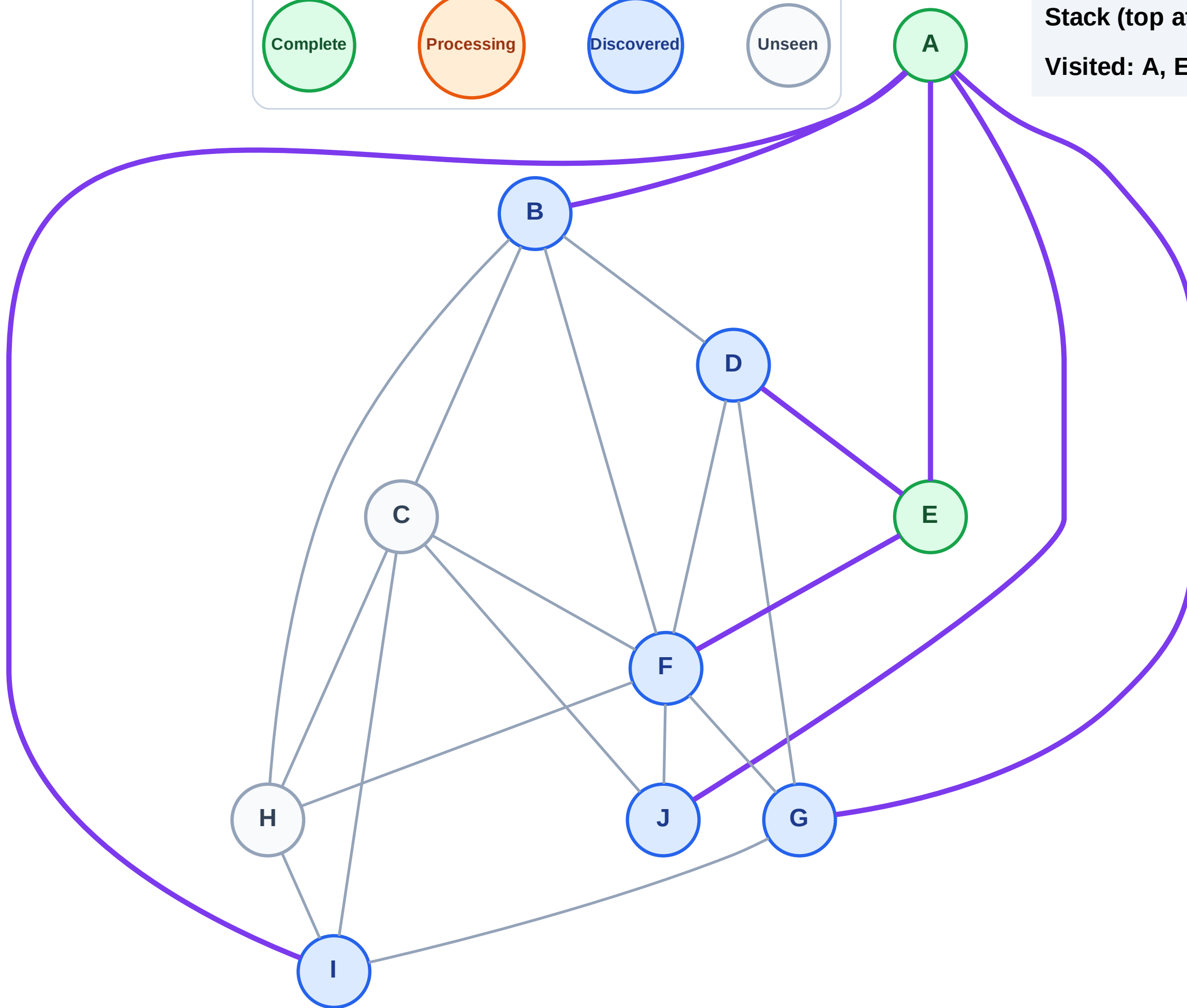
Step 5: Discover J, I, G, B from A

Legend



Stack (top at right): F | D | J | I | G | B

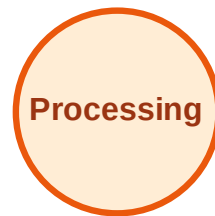
Visited: A, E



DFS Traversal

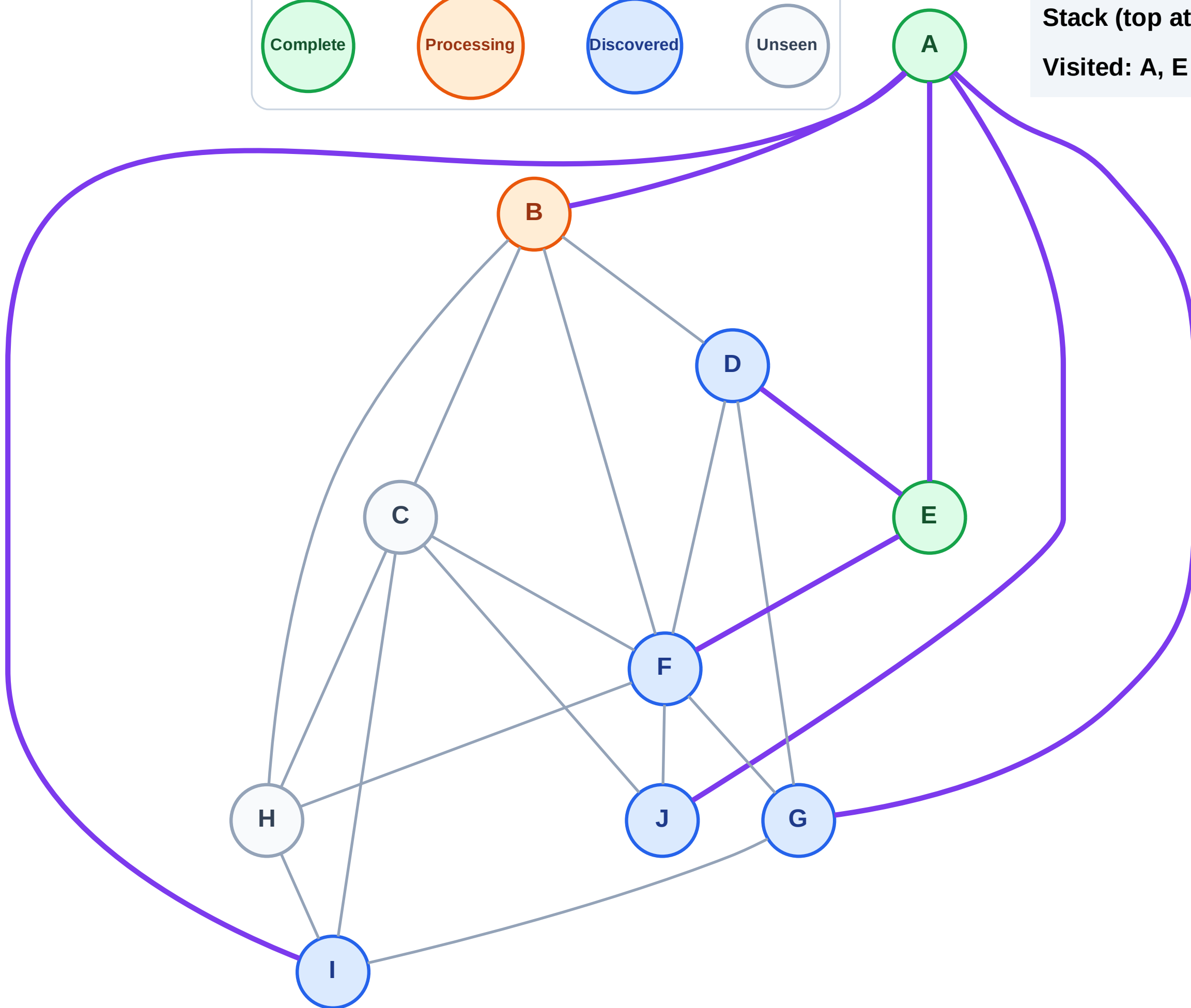
Step 6: Process node B

Legend



Stack (top at right): F | D | J | I | G

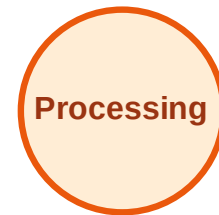
Visited: A, E



DFS Traversal

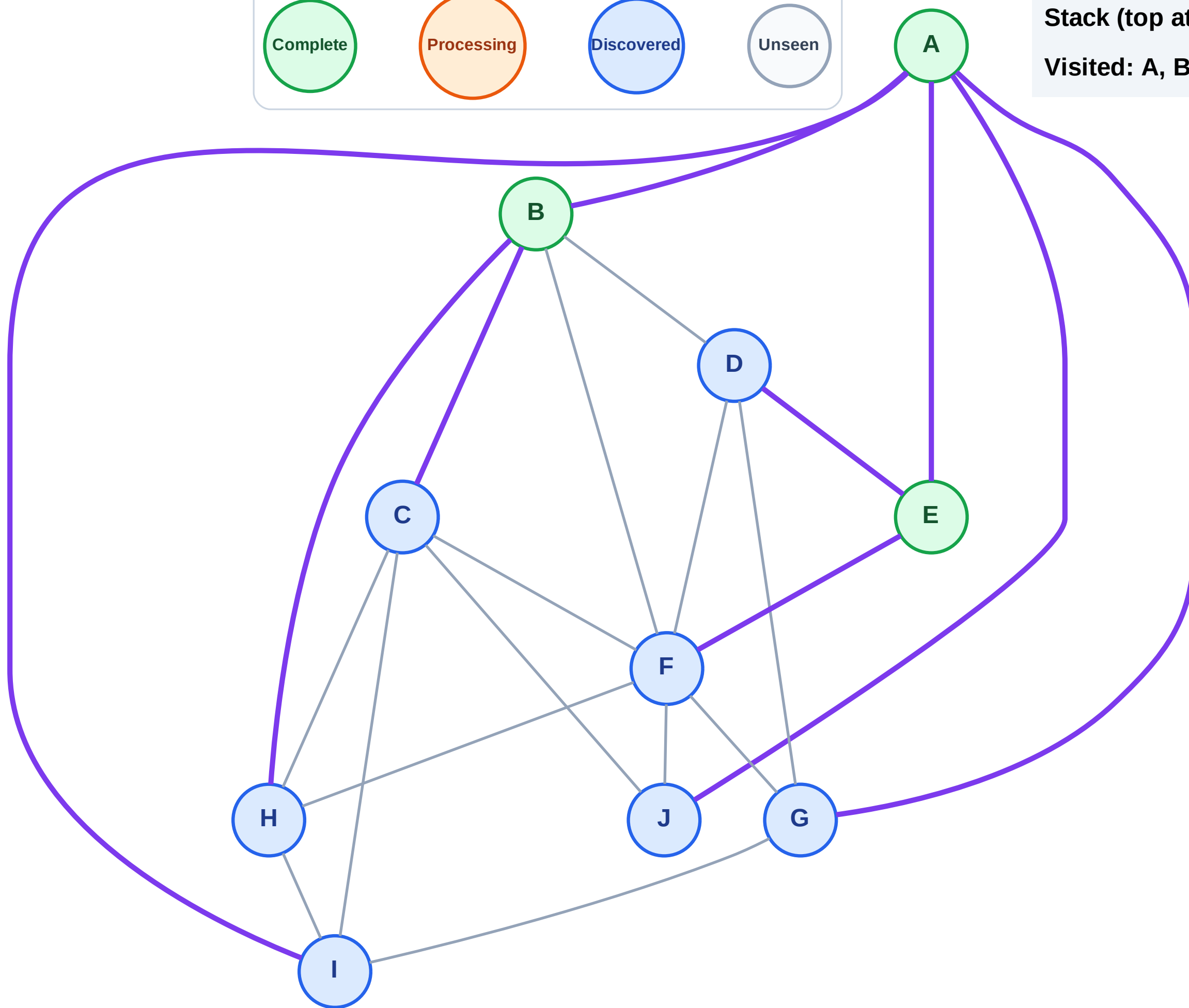
Step 7: Discover H, C from B

Legend



Stack (top at right): F | D | J | I | G | H | C

Visited: A, B, E



DFS Traversal

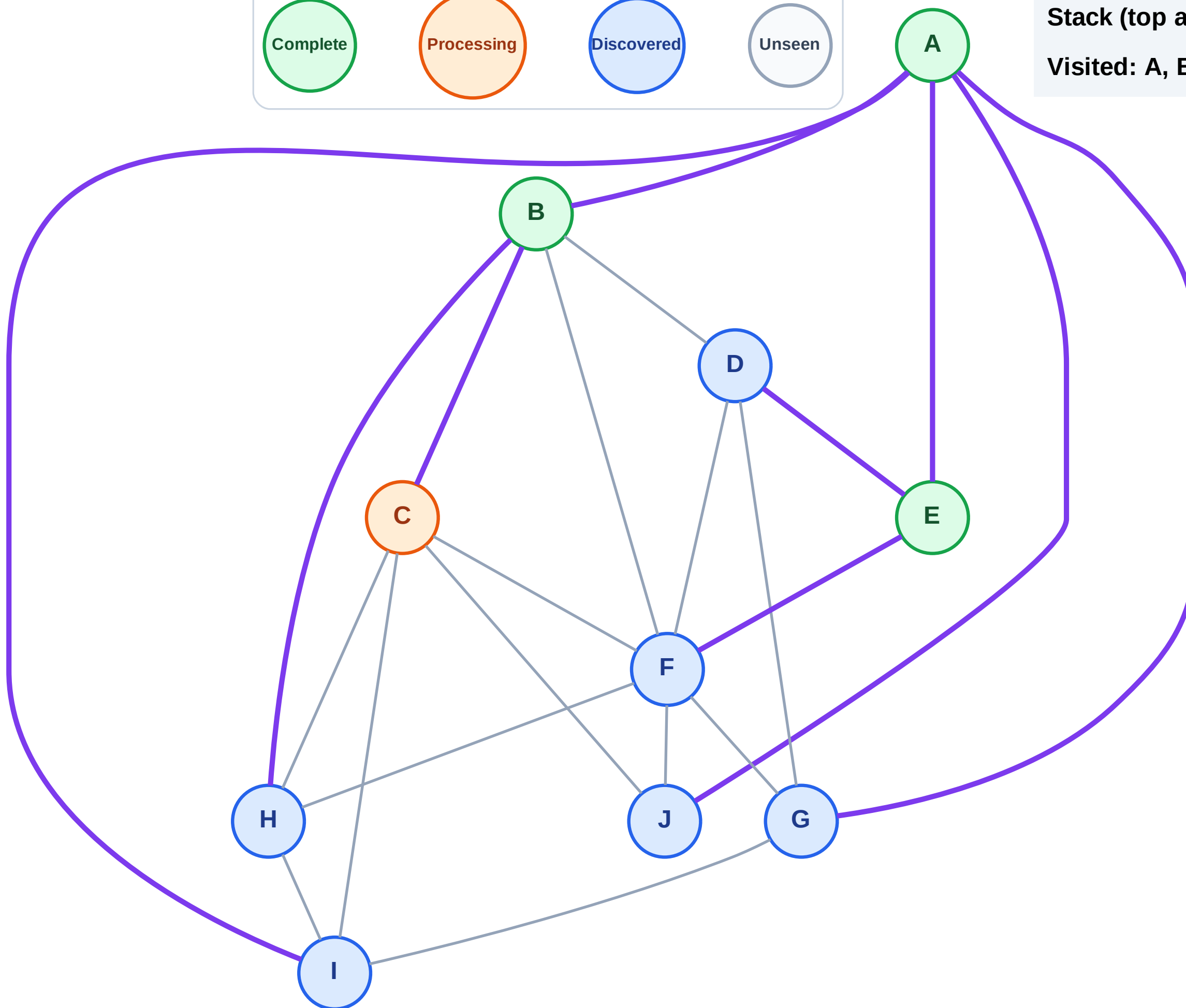
Step 8: Process node C

Legend



Stack (top at right): F | D | J | I | G | H

Visited: A, B, E



DFS Traversal

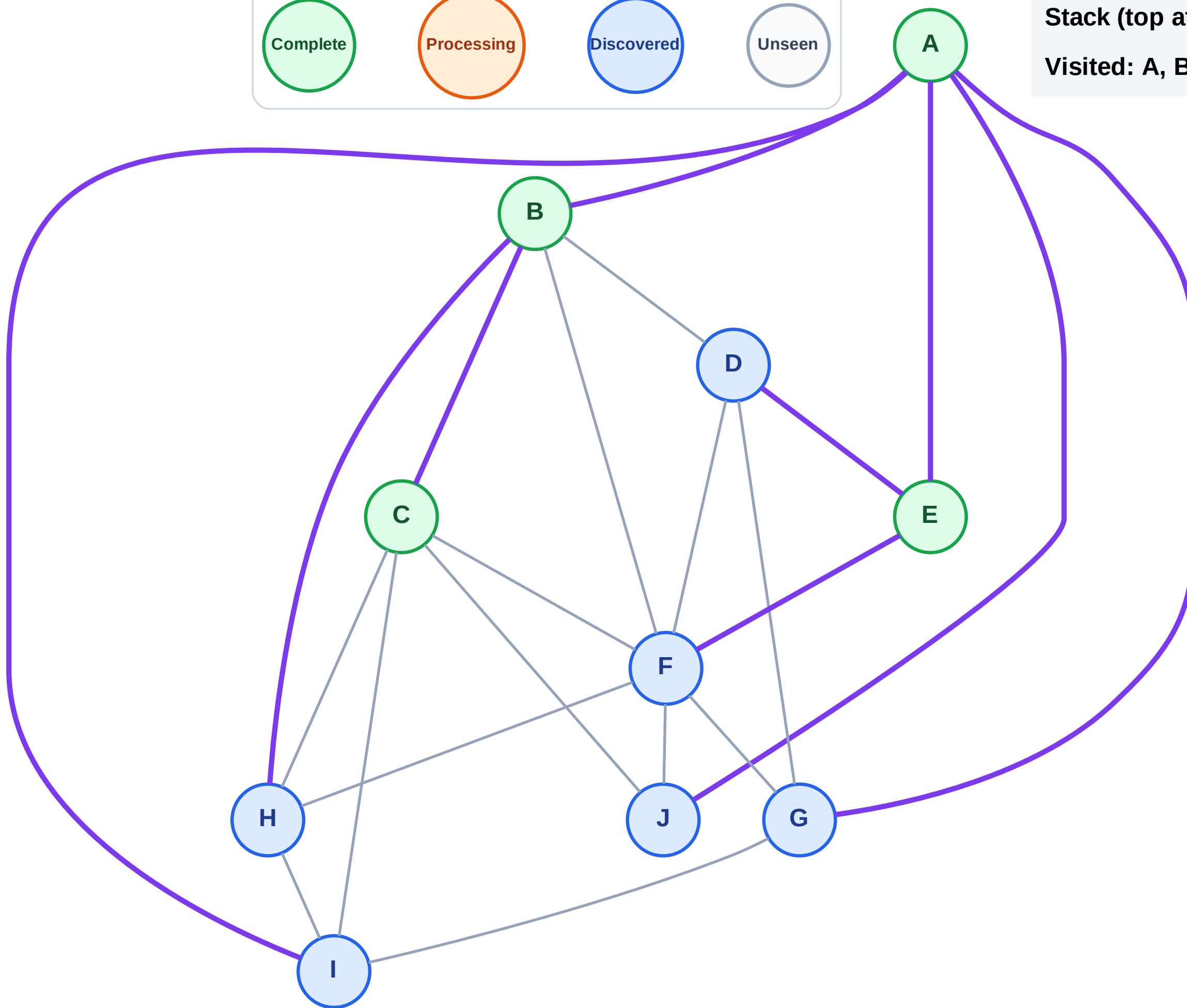
Step 9: No new neighbors from C

Legend



Stack (top at right): F | D | J | I | G | H

Visited: A, B, C, E



DFS Traversal

Step 10: Process node H

Legend

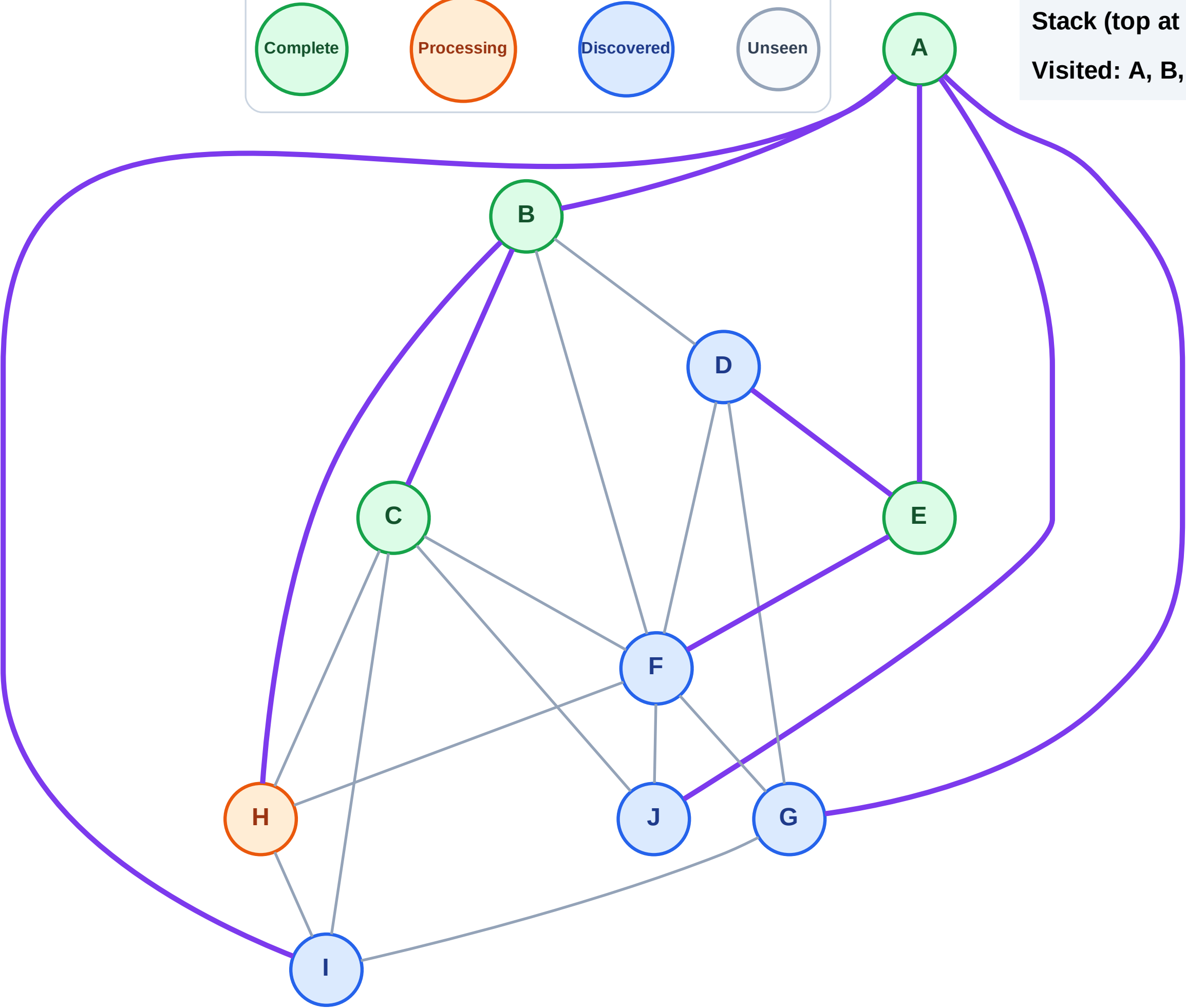
Complete

Processing

Discovered

Unseen

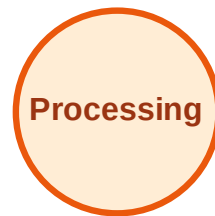
Stack (top at right): F | D | J | I | G
Visited: A, B, C, E



DFS Traversal

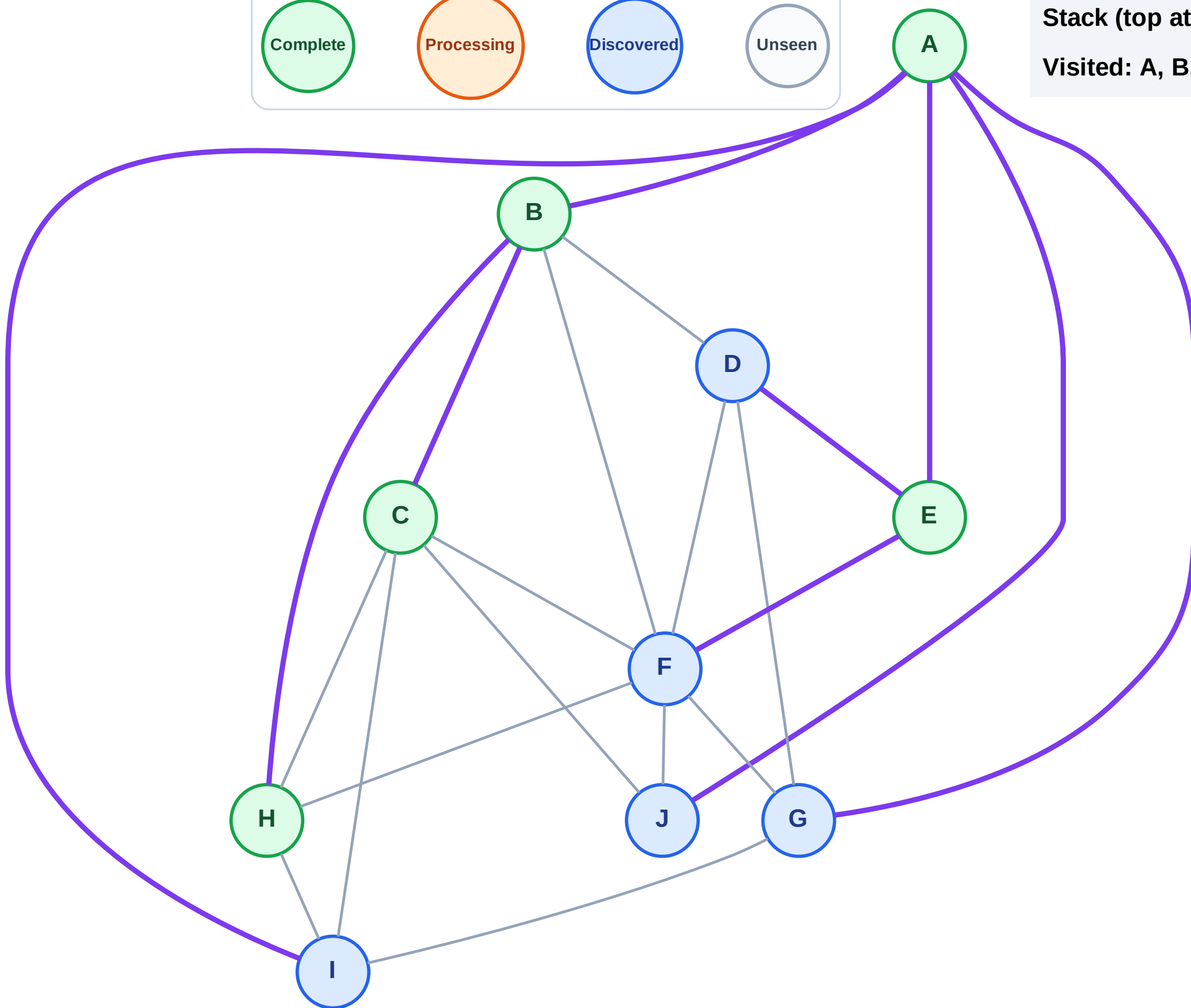
Step 11: No new neighbors from H

Legend



Stack (top at right): F | D | J | I | G

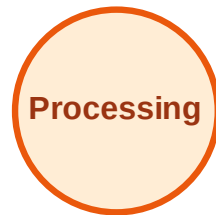
Visited: A, B, C, E, H



DFS Traversal

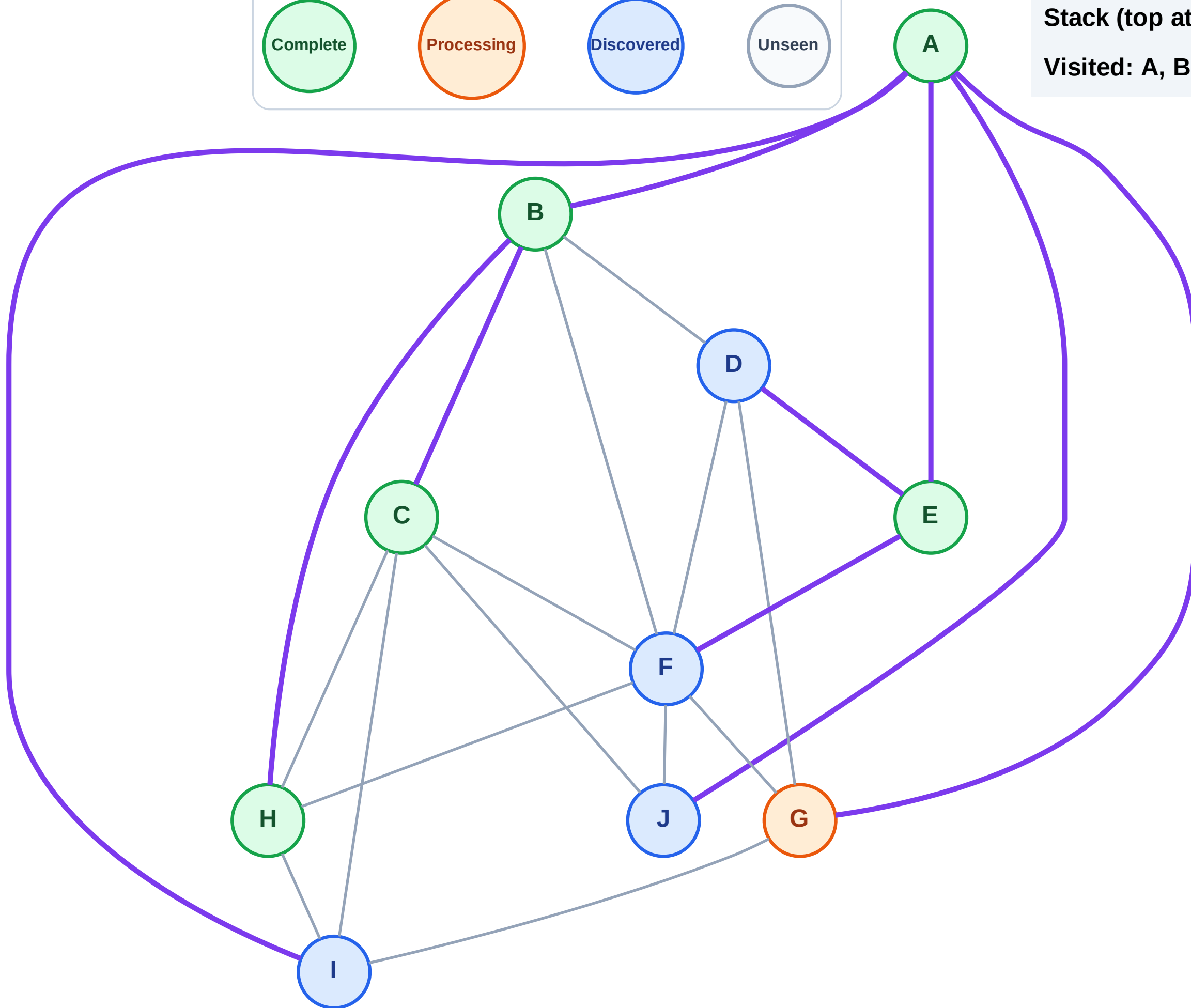
Step 12: Process node G

Legend



Stack (top at right): F | D | J | I

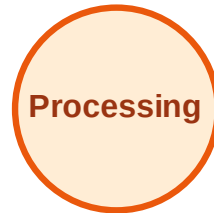
Visited: A, B, C, E, H



DFS Traversal

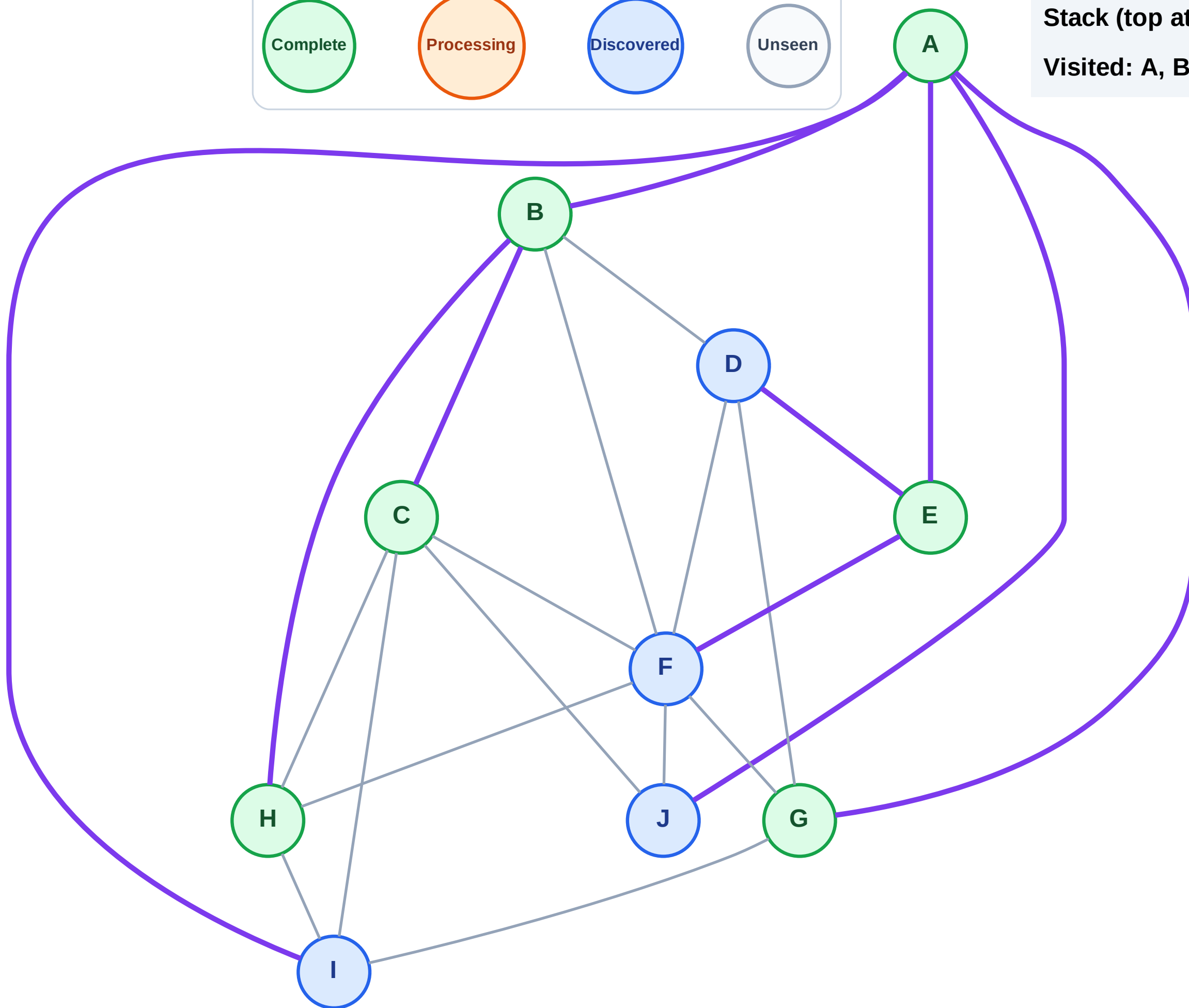
Step 13: No new neighbors from G

Legend



Stack (top at right): F | D | J | I

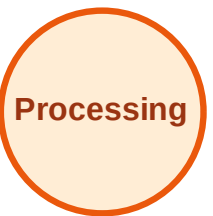
Visited: A, B, C, E, G, H



DFS Traversal

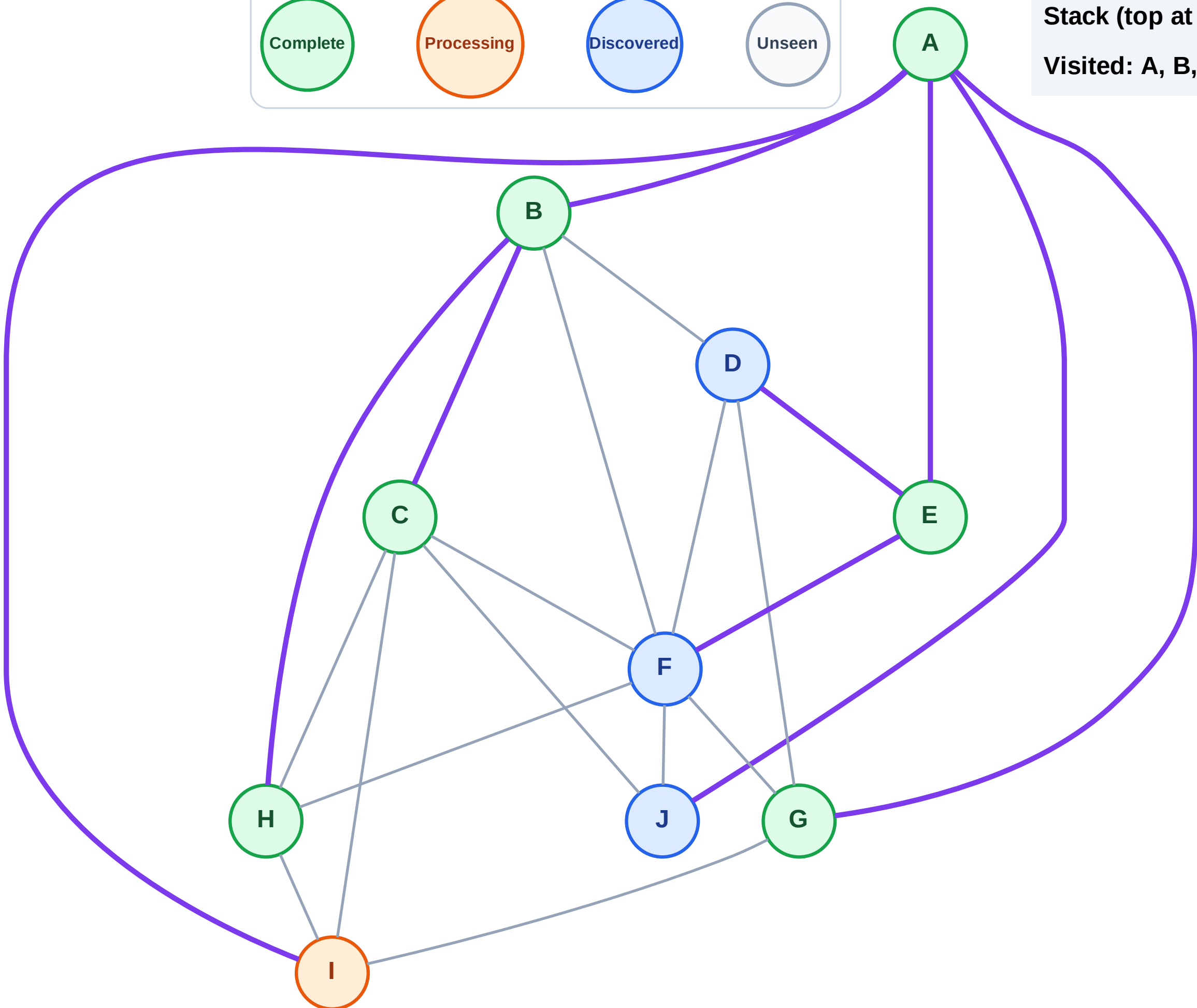
Step 14: Process node I

Legend



Stack (top at right): F | D | J

Visited: A, B, C, E, G, H



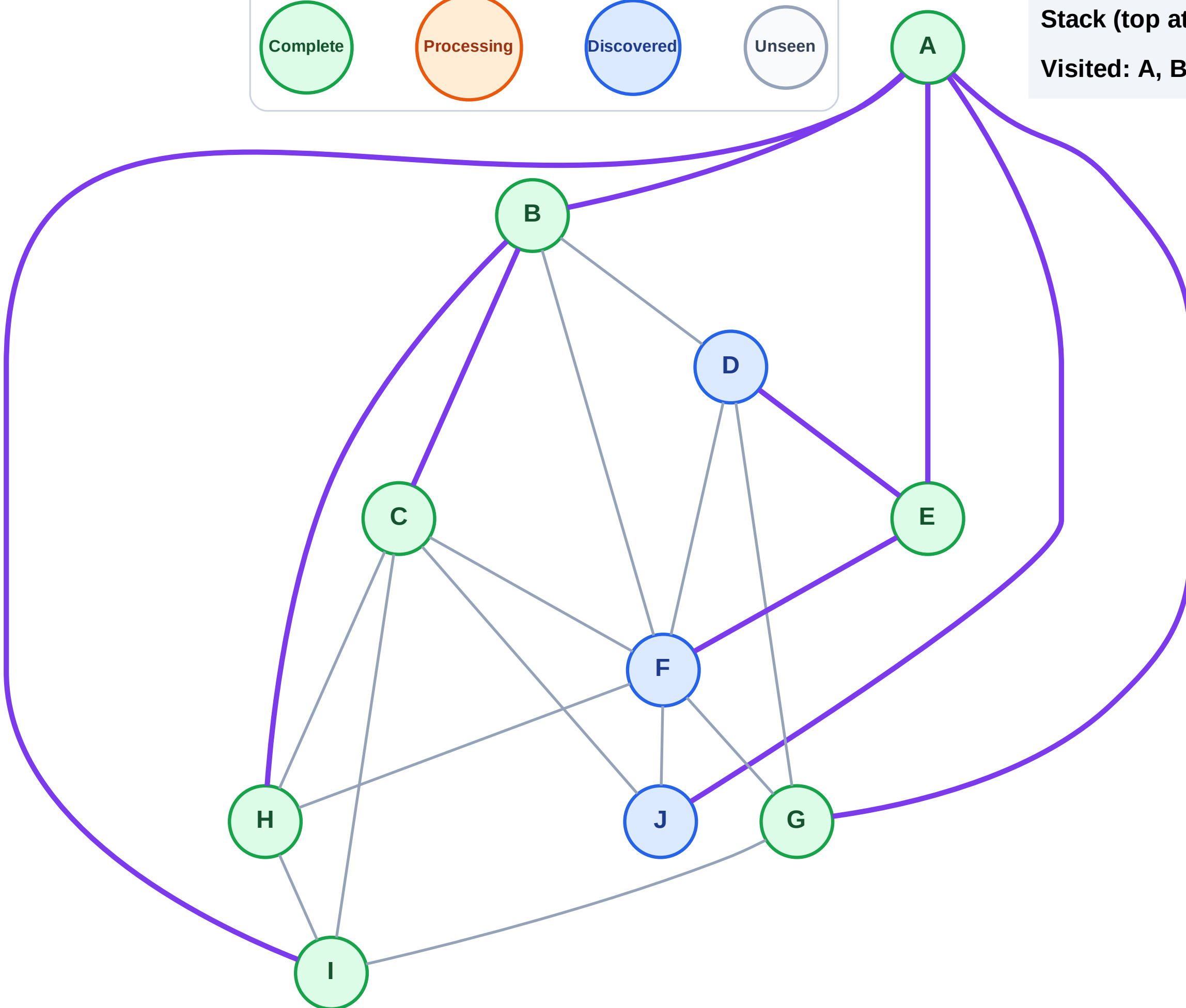
Step 15: No new neighbors from I

Grand

The diagram consists of four circles arranged horizontally, each containing a label. From left to right: a green circle with the word 'Complete', an orange circle with the word 'Processing', a blue circle with the word 'Discovered', and a light blue circle with the word 'Unseen'. The circles are connected by a light blue line that curves around the bottom.

Unseen

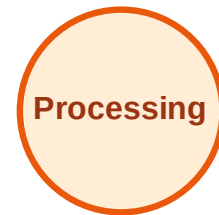
Visited: A, B, C, E, G, H, I



DFS Traversal

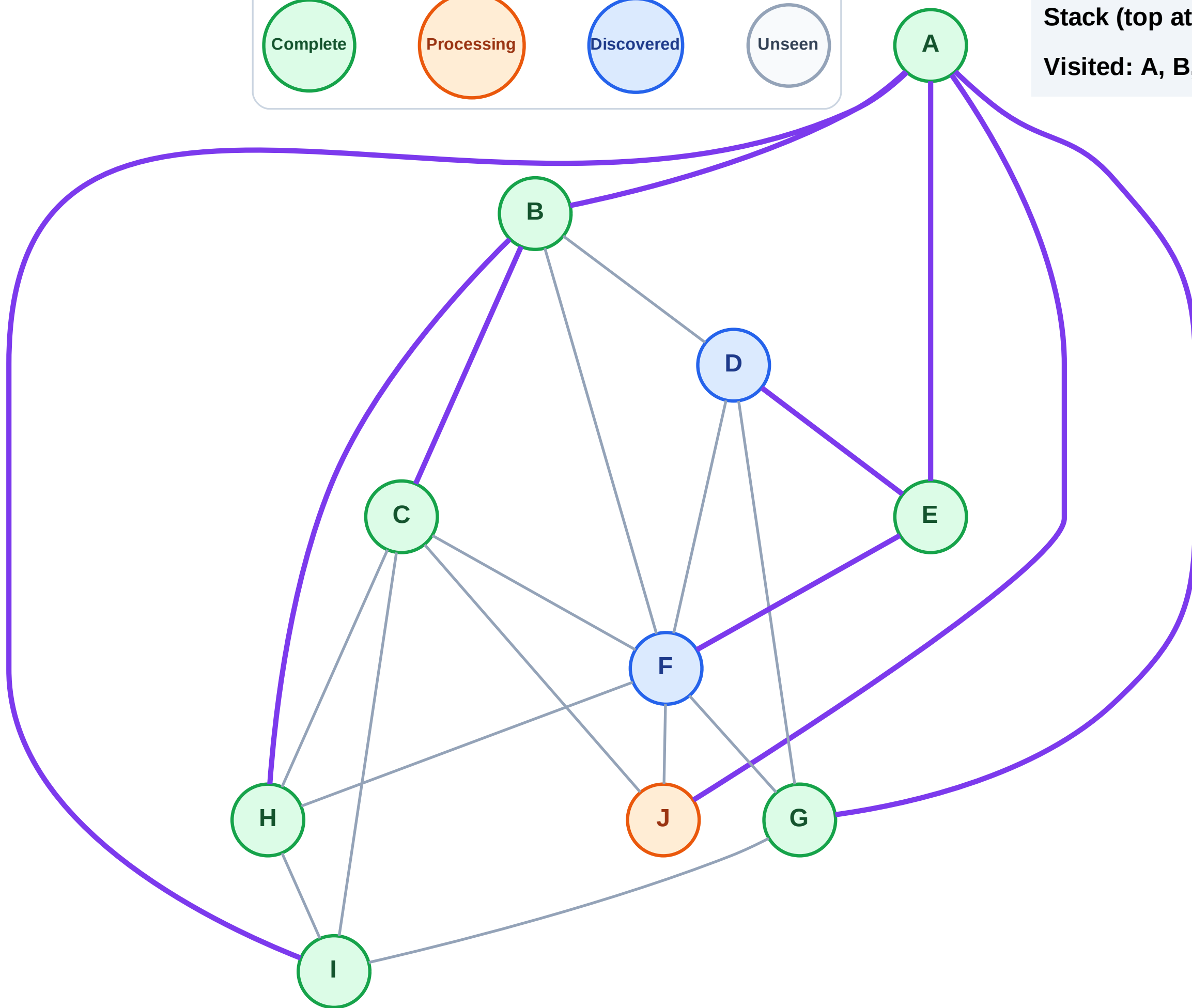
Step 16: Process node J

Legend



Stack (top at right): F | D

Visited: A, B, C, E, G, H, I



DFS Traversal

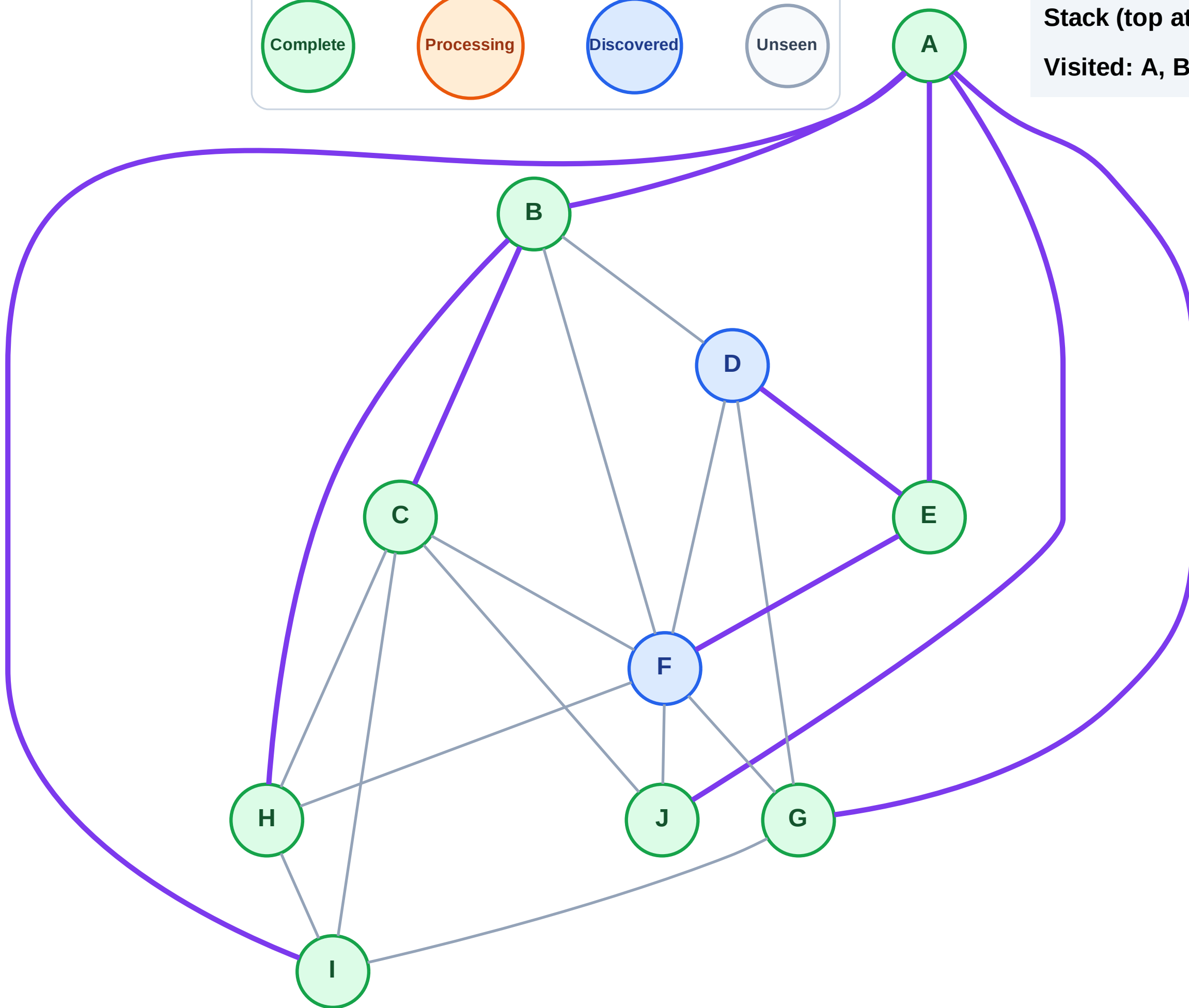
Step 17: No new neighbors from J

Legend



Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



DFS Traversal

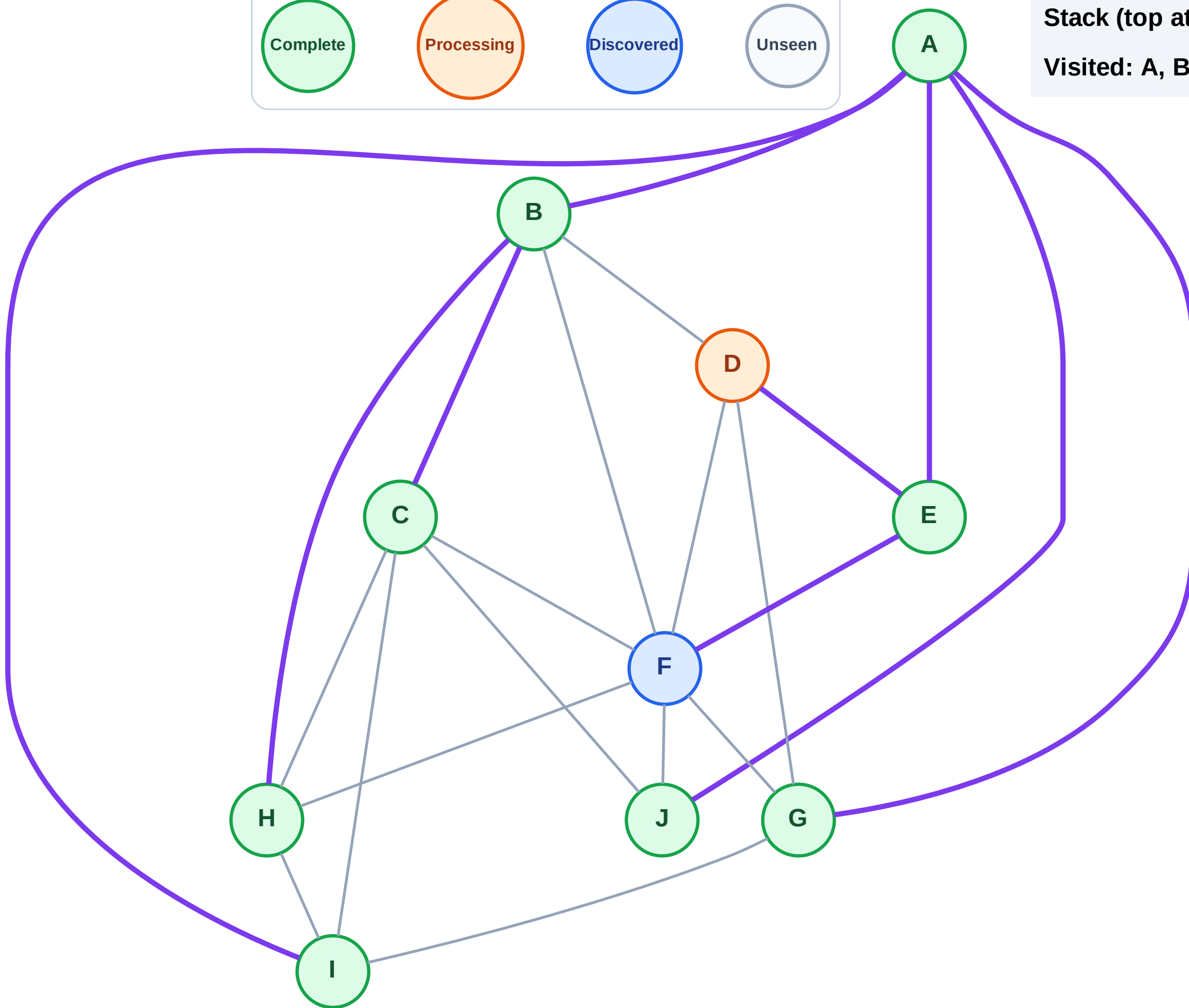
Step 18: Process node D

Legend



Stack (top at right): F

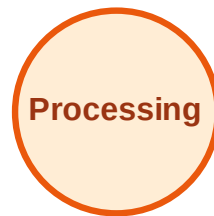
Visited: A, B, C, E, G, H, I, J



DFS Traversal

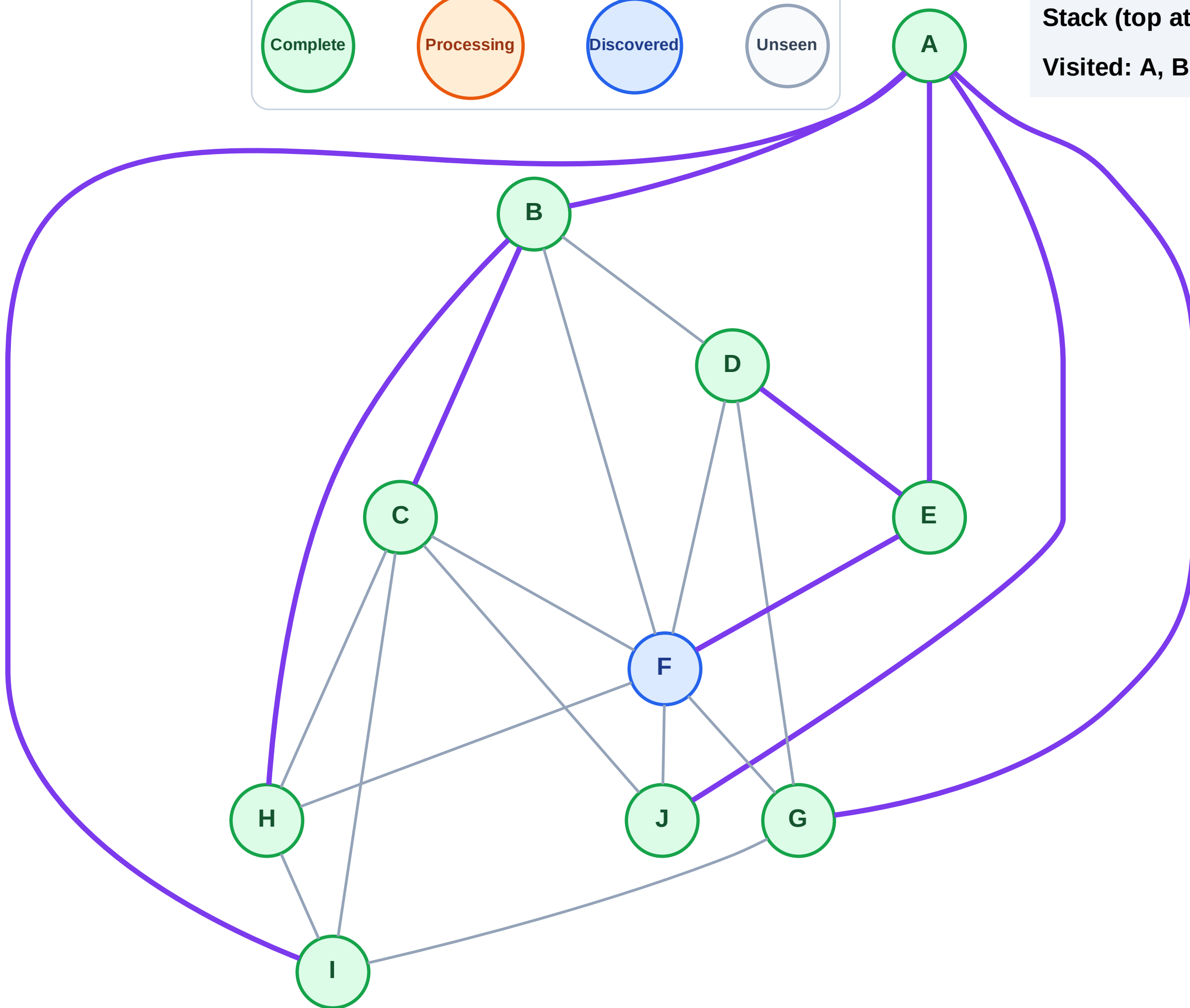
Step 19: No new neighbors from D

Legend



Stack (top at right): F

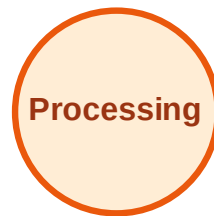
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

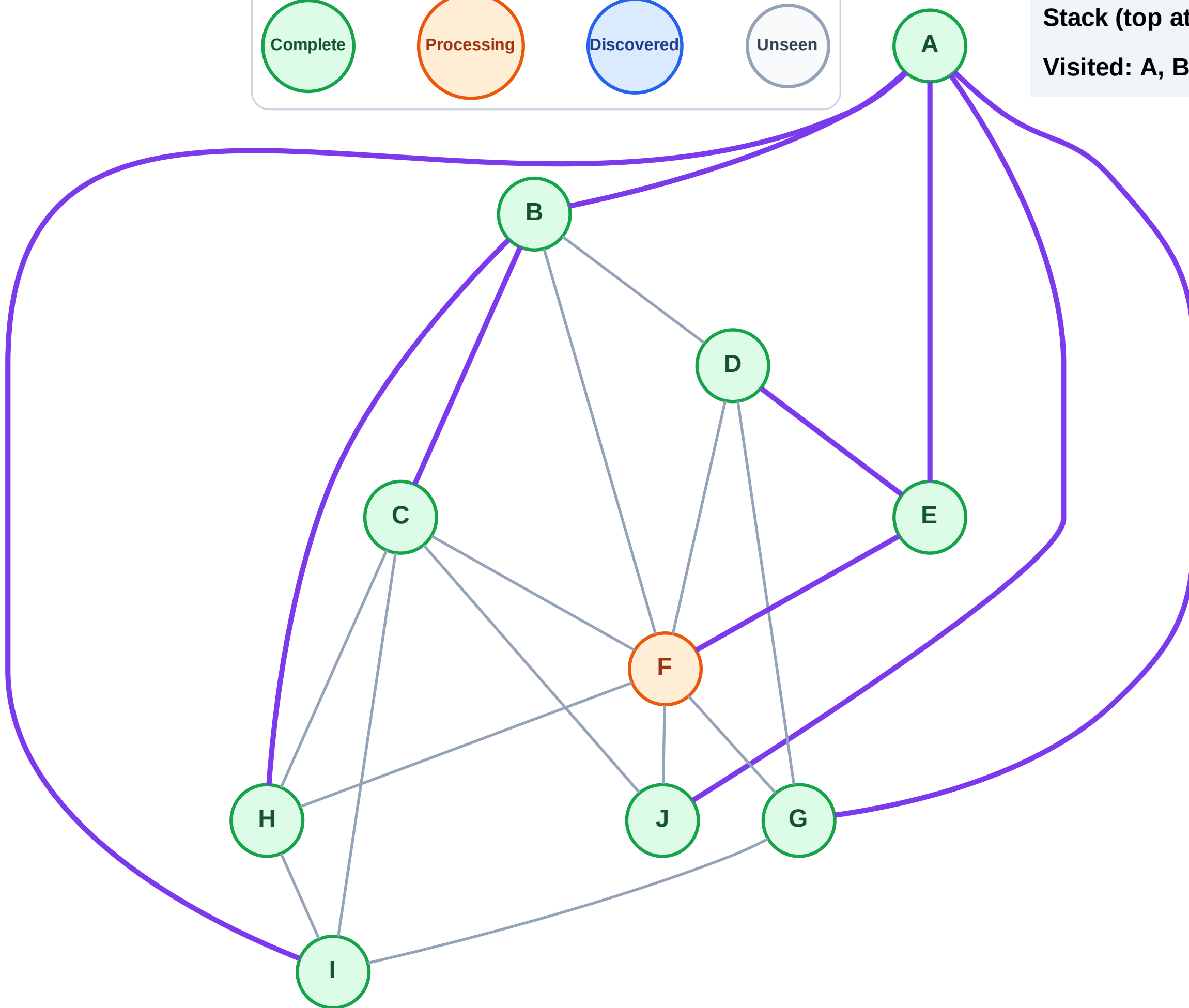
Step 20: Process node F

Legend



Stack (top at right): (empty)

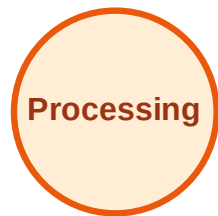
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

